

# Pseudotraces on Almost Unital and Finite-Dimensional Algebras

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## Abstract

We introduce the notion of almost unital and finite-dimensional (AUF) algebras, which are associative  $\mathbb{C}$ -algebras that may be non-unital or infinite-dimensional, but have sufficiently many idempotents. We show that the pseudotrace construction, originally introduced by Hattori and Stallings for unital finite-dimensional algebras, can be generalized to AUF algebras.

Let  $A$  be an AUF algebra. Suppose that  $G$  is a projective generator in the category  $\text{Coh}_L(A)$  of finitely generated left  $A$ -modules that are quotients of free left  $A$ -modules, and let  $B = \text{End}_{A,-}(G)^{\text{op}}$ . We prove that the pseudotrace construction yields an isomorphism between the spaces of symmetric linear functionals  $\text{SLF}(A) \xrightarrow{\cong} \text{SLF}(B)$ , and that the non-degeneracies on the two sides are equivalent.

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## 0 Introduction

In [Miy04], Miyamoto introduced the pseudo- $q$ -trace construction for modules of vertex operator algebras (VOAs), generalizing the usual  $q$ -trace. His primary motivation was to address the failure of modular invariance for  $q$ -traces in the case of  $C_2$ -cofinite but irrational VOAs. While Zhu's theorem in [Zhu96] establishes modular invariance for  $q$ -traces in the rational setting, this result does not extend to the irrational case—unless  $q$ -traces are replaced with pseudo- $q$ -traces.

Miyamoto's original approach is quite involved. Moreover, his dimension formula for the space of torus conformal blocks is expressed in terms of higher Zhu algebras. This presents two drawbacks: first, higher Zhu algebras are difficult to compute in practice; second, their connection to the VOA module category is not transparent.

Later, Arike [Ari10] and Arike-Nagatomo [AN13] introduced a simplified version of the pseudo- $q$ -trace construction based on the idea of Hattori [Hat65] and Stallings [Sta65]. Below, we briefly outline this approach.

Let  $A$  be an algebra, and let  $B$  be a unital finite-dimensional algebra. Let  $M$  be a finite-dimensional  $A$ - $B$  bimodule, projective as a right  $B$ -module. By the projectivity, there is a (finite) left coordinate system of  $M$ , namely, elements  $\alpha_1, \dots, \alpha_n \in \text{Hom}_B(B, M)$  and  $\check{\alpha}^1, \dots, \check{\alpha}^n \in \text{Hom}_B(M, B)$  satisfying  $\sum_i \alpha_i \circ \check{\alpha}^i = \text{id}_M$ . Then the linear map

$$A \rightarrow B \quad x \mapsto \sum_i \check{\alpha}^i \circ x \circ \alpha_i(1_B)$$

descends to a linear map  $A/[A, A] \rightarrow B/[B, B]$  which is independent of the choice of the left coordinate system. Its pullback gives a linear map

$$\text{SLF}(B) \rightarrow \text{SLF}(A) \quad \phi \mapsto \text{Tr}^\phi \tag{0.1}$$

where  $\text{SLF}(A)$  is the space of symmetric linear functionals on  $A$ —that is, linear maps  $\psi : A \rightarrow \mathbb{C}$  satisfying  $\psi(xy) = \psi(yx)$  for all  $x, y \in A$ —and  $\text{SLF}(B)$  is the space of symmetric linear functionals on  $B$ . The above map is called the **pseudotrace construction**. Note that a typical choice of  $A$  is  $\text{End}_B(M)$ .

The pseudotrace construction is applied to the VOA setting as follows. Let  $\mathbb{V}$  be an  $\mathbb{N}$ -graded  $C_2$ -cofinite VOA with central charge  $c$ , and let  $\mathbb{M}$  be a grading-restricted generalized  $\mathbb{V}$ -module. Then  $\mathbb{M}$  admits a decomposition  $\mathbb{M} = \bigoplus_{\lambda \in \mathbb{C}} \mathbb{M}_{[\lambda]}$  into generalized eigenspaces of  $L(0)$ , where each  $\mathbb{M}_{[\lambda]}$  is finite-dimensional. Let  $\text{End}_{\mathbb{V}}(\mathbb{M})$  be the algebra of linear operators on  $\mathbb{M}$  commuting with the action of  $\mathbb{V}$ , which is necessarily unital and finite-dimensional. Let  $B$  be a unital subalgebra of  $\text{End}_{\mathbb{V}}(\mathbb{M})^{\text{op}}$ . Assume that  $\mathbb{M}$  is a projective right  $B$ -module, equivalently, each  $\mathbb{M}_{[\lambda]}$  is  $B$ -projective. Let  $\phi \in \text{SLF}(B)$ . Then for  $v \in \mathbb{V}$ , the expression

$$\text{Tr}^\phi(Y_{\mathbb{M}}(v, z)q^{L(0) - \frac{c}{24}}) = \sum_{\lambda \in \mathbb{C}} \text{Tr}^\phi(P(\lambda)Y_{\mathbb{M}}(v, z)q^{L(0) - \frac{c}{24}}P(\lambda)) \tag{0.2}$$

converges absolutely for  $z \in \mathbb{C}$  and  $0 < |q| < 1$ , and defines a torus conformal block. Here,  $P(\lambda)$  is the projection of  $\overline{\mathbb{M}} := \prod_{\mu \in \mathbb{C}} \mathbb{M}_{[\mu]}$  onto  $\mathbb{M}_{[\lambda]}$ . Then each  $P(\lambda)Y_{\mathbb{M}}(v, z)q^{L(0) - \frac{c}{24}}P(\lambda)$

is a linear operator on  $\mathbb{M}_{[\lambda]}$  commuting with the right action of  $B$ , and hence  $\mathrm{Tr}^\phi$  can be defined on it.

Based on this formulation, in [GR19, Conjecture 5.8], Gainutdinov and Runkel proposed a conjecture that directly relates the space of torus conformal blocks of a  $C_2$ -cofinite VOA  $\mathbb{V}$  to the linear structure of the category  $\mathrm{Mod}(\mathbb{V})$  of grading-restricted generalized  $\mathbb{V}$ -modules. Let  $\mathbb{G}$  be a projective generator in  $\mathrm{Mod}(\mathbb{V})$ , and let  $B = \mathrm{End}_{\mathbb{V}}(\mathbb{G})$ . Then  $\mathbb{G}$  is  $B$ -projective. The conjecture asserts that the linear map sending each  $\phi \in \mathrm{SLF}(B)$  to (0.2) defines an isomorphism between  $\mathrm{SLF}(B)$  and the space of torus conformal blocks of  $\mathbb{V}$ .

The purpose of this note is to establish results in the theory of associative algebras that are essential for proving the Gainutdinov-Runkel conjecture. The actual resolution of the conjecture will appear in the forthcoming paper [GZ25].

Our approach stems from recognizing a structural analogy between the Gainutdinov-Runkel conjecture and a classical result in associative algebra: If  $A$  is a unital finite-dimensional algebra and  $M$  is a projective generator in the category of finite-dimensional left  $A$ -modules, then  $M$  is projective over  $B := \mathrm{End}_A(M)^{\mathrm{op}}$ , and the pseudotracer map (0.1) is a linear isomorphism. This result was suggested in [BBG21, Sec. 2] and was proved in [Ari10] in the special case that  $M = Ae$  where  $e$  is a basic idempotent.

However, this classical result is not directly applicable to the Gainutdinov-Runkel conjecture. We need to generalize it to a larger class of associative algebras than unital finite-dimensional ones. In particular, we must consider infinite-dimensional algebras that can be approximated, in a certain sense, by finite-dimensional (and possibly unital) algebras. The need to consider infinite-dimensional associative algebras in the study of irrational VOAs has also been recognized in recent years from different perspectives, such as Huang's associative algebra  $A^\infty(\mathbb{V})$  introduced in [Hua24], and the mode transition algebra introduced by Damiolini-Gibney-Krashen in [DGK25].

The infinite-dimensional algebra required for the proof of the Gainutdinov-Runkel conjecture is different from the above mentioned algebras. In [GZ25], we will show that the end

$$\mathbb{E} := \int_{M \in \mathrm{Mod}(\mathbb{V})} M \otimes_{\mathbb{C}} M'$$

a priori an object of  $\mathrm{Mod}(\mathbb{V}^{\otimes 2})$ , carries a structure of an associative  $\mathbb{C}$ -algebra that is compatible with its  $\mathbb{V}^{\otimes 2}$ -module structure. This algebra  $\mathbb{E}$  is an example of an **almost unital and finite-dimensional algebra**<sup>1</sup> (abbreviated as **AUF algebra**), meaning that  $\mathbb{E}$  has a collection of mutually orthogonal idempotents  $(e_i)_{i \in \mathcal{I}}$  such that  $\mathbb{E} = \sum_{i,j \in \mathcal{I}} e_i \mathbb{E} e_j$  where each summand  $e_i \mathbb{E} e_j$  is finite-dimensional. (This sum is automatically direct.) In fact,  $\mathbb{E}$  has only finitely many irreducibles. We call such an algebra **strongly AUF**.

The main result of this note is a generalization of the aforementioned isomorphism between spaces of symmetric linear functionals to the setting of strongly AUF algebras. More precisely, we prove that the pseudotracer construction defines a linear isomorphism  $\mathrm{SLF}(B) \simeq \mathrm{SLF}(A)$  where  $A$  is strongly AUF,  $M$  is a projective generator of the category  $\mathrm{Coh}_L(A)$  of **coherent left  $A$ -modules** (i.e., finitely generated left  $A$ -modules that are quotients of free ones), and  $B = \mathrm{End}_A(M)^{\mathrm{op}}$ . See Thm. 9.4. Moreover, we show that

<sup>1</sup>Here, “almost” modifies the entire phrase “unital and finite-dimensional”, not just “unital”.

the symmetric linear functional on  $B$  is non-degenerate if and only if the corresponding functional on  $A$  is non-degenerate. See Thm. 10.4.

Since the associative algebra structure on the end  $\mathbb{E}$  will not be developed in this note, we present some alternative examples of AUF algebras for illustration. Let  $U(\mathbb{V})$  be the universal algebra of  $\mathbb{V}$  as defined in [FZ92]. Let

$$U(\mathbb{V})^{\text{reg}} = \bigoplus_{\lambda, \mu \in \mathbb{C}} U(\mathbb{V})_{[\lambda, \mu]}$$

where  $U(\mathbb{V})_{[\lambda, \mu]}$  is the subspace of joint generalized-eigenvectors of the left and right actions of  $L(0)$  corresponding to the eigenvalues  $\lambda$  and  $\mu$  respectively. The following properties are shown in [MNT10]: Each  $U(\mathbb{V})_{[\lambda, \mu]}$  is finite-dimensional. For each  $\lambda, \mu, \nu \in \mathbb{C}$  one has

$$U(\mathbb{V})_{[\lambda, \mu]} U(\mathbb{V})_{[\mu, \nu]} \subset U(\mathbb{V})_{[\lambda, \nu]}$$

In particular,  $U(\mathbb{V})^{\text{reg}}$  is a subalgebra of  $U(\mathbb{V})$ . Moreover, there is an increasing sequence of idempotents  $(1_n)_{n \in \mathbb{Z}_+}$  such that  $U(\mathbb{V})^{\text{reg}} = \bigcup_n 1_n U(\mathbb{V})^{\text{reg}} 1_n$ . (See [MNT10, Sec. 2.6].) Therefore,  $U(\mathbb{V})^{\text{reg}}$  is AUF, since the family of orthogonal idempotents in the definition of AUF algebras can be chosen to be  $(1_{n+1} - 1_n)_{n \in \mathbb{Z}_+}$ .

For a more elementary and concrete example, consider the following. Let  $B$  be a unital finite-dimensional algebra. Let  $M$  be a right  $B$ -module. Equip  $M$  with a grading

$$M = \bigoplus_{i \in \mathcal{I}} M(i)$$

where each  $M(i)$  is finite-dimensional and is preserved by the right action of  $B$ . Let  $A$  be

$$\begin{aligned} \text{End}_B^0(M) &:= \{T \in \text{End}(M) : (Tm)b = T(mb) \text{ for all } m \in M, b \in B, \\ &\quad T|_{M(i)} = 0 \text{ for all but finitely many } i \in \mathcal{I}\} \end{aligned}$$

Then  $A$  is clearly an AUF algebra, with the family of mutually orthogonal idempotents given by the projections  $e_i$  of  $M$  onto  $M(i)$ .

In fact, any strongly AUF algebra arises from such a construction. More precisely, an algebra is strongly AUF if and only if it is isomorphic to some  $\text{End}_B^0(M)$ , where  $M$  and  $B$  satisfy the above conditions and, in addition,  $M$  is a projective generator in the category of right  $B$ -modules. See Thm. 11.9.

Note that the relationship between  $\text{End}_B^0(M)$  and  $C_2$ -cofinite VOAs is straightforward: If  $\mathbb{M} \in \text{Mod}(\mathbb{V})$  is equipped with the grading  $\bigoplus_{\lambda \in \mathbb{C}} \mathbb{M}_{[\lambda]}$  given by the generalized eigenspaces of  $L(0)$ , and if  $B$  is a unital subalgebra of  $\text{End}_{\mathbb{V}}(\mathbb{M})^{\text{op}}$  such that  $\mathbb{M}$  is projective as a right  $B$ -module, then each  $P(\lambda)Y_{\mathbb{M}}(v, z)q^{L(0) - \frac{c}{24}}P(\lambda)$  appearing in (0.2) lies in  $\text{End}_B^0(\mathbb{M})$ . Therefore, the main result of this note on pseudotraces (Thm. 10.4) can be applied to  $C_2$ -cofinite VOAs. Details of this application will be presented in [GZ25].

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# 1 Preliminaries

Throughout this note, algebras are associative, not necessarily unital, and over  $\mathbb{C}$ . Let  $\mathbb{N} = \{0, 1, 2, \dots\}$  and  $\mathbb{Z}_+ = \{1, 2, \dots\}$ . For any vector spaces  $V, W$ , we let  $\text{Hom}(V, W) = \text{Hom}_{\mathbb{C}}(V, W)$  be the space of linear maps  $V \rightarrow W$ , and let  $\text{End}(V) = \text{Hom}(V, V)$ .

Let  $A$  be an algebra. Its opposite algebra is denoted by  $A^{\text{op}}$ . If  $M, N$  are left (resp. right)  $A$ -modules, we let  $\text{Hom}_{A,-}(M, N)$  (resp.  $\text{Hom}_{-,A}(M, N)$ ) be the space of linear maps  $M \rightarrow N$  intertwining the left (resp. right) actions of  $A$ .

An **idempotent**  $e \in A$  is an element satisfying  $e^2 = e$ . If  $e, f \in A$  are idempotents, we write  $e \leq f$  if  $ef = fe = e$ . Equivalently,  $f = e + e'$  where  $e' \in A$  is an idempotent **orthogonal** to  $e$  (i.e.  $ee' = e'e = 0$ ). We call  $e$  a **sub-idempotent** of  $f$  if  $e \leq f$ . We say that a nonzero idempotent  $e$  is **primitive** if the only idempotent  $f$  satisfying  $f \leq e$  is  $f = 0$  and  $f = e$ .

In this section, we review some well-known facts about associative algebras. Since, unlike many references, our algebras are not assumed to be unital, we include proofs for the reader's convenience.

**Definition 1.1.** Let  $u, v \in A$ . We say that  $(u, v)$  is a pair of **partial isometries** in  $A$  if the following are true:

- (a)  $p := vu$  and  $q := uv$  are idempotents.
- (b)  $u \in qAp$  and  $v \in pAq$ .

In this case, we also say that  $u$  is a partial isometry from  $p$  to  $q$ , and that  $v$  is a partial isometry from  $q$  to  $p$ . We say that two idempotents are **equivalent** if there are partial isometries between them.

**Proposition 1.2.** Let  $e, f \in A$  be idempotents. Then an element of  $\text{Hom}_{A,-}(Ae, Af)$  is precisely the right multiplication of an element of  $eAf$ . In particular, we have an algebra isomorphism

$$\text{End}_{A,-}(Ae)^{\text{op}} \simeq eAe$$

*Proof.* Clearly the right multiplication by some element of  $eAf$  yields an element of  $\text{Hom}_{A,-}(Ae, Af)$ . Conversely, suppose that  $T \in \text{Hom}_{A,-}(Ae, Af)$ . Let  $x = T(e)$ , which belongs to  $Af$ . Since  $ex = eT(e) = T(ee) = T(e) = x$ , we see that  $x \in eAf$ . For each  $y \in A$ , we have  $T(ye) = yT(e) = yx = yex$ , which shows that  $T$  is the right multiplication by  $x$ .  $\square$

**Corollary 1.3.** Let  $e, f$  be idempotents in  $A$ . The following are equivalent:

- (1)  $Ae \simeq Af$  as left  $A$ -modules.
- (2) There is a partial isometry from  $e$  to  $f$ .

*Proof.* (1) $\Rightarrow$ (2): Let  $T \in \text{Hom}_{A,-}(Ae, Af)$  be an isomorphism with inverse  $T^{-1} \in \text{Hom}_{A,-}(Af, Ae)$ . By Prop. 1.2,  $T$  and  $T^{-1}$  are realized by the right multiplications of  $u \in eAf$  and  $v \in fAe$  respectively. Since  $TT^{-1} = 1_{Af}$ , we have  $vu = f$ . Since  $T^{-1}T = 1_{Ae}$ , we have  $uv = e$ .

(2) $\Rightarrow$ (1): Let  $u \in eAf$  and  $v \in fAe$  such that  $uv = e, vu = f$ . Then the right multiplication of  $u$  on  $Ae$  has inverse being the right multiplication of  $v$ . So  $Ae \simeq Af$ .  $\square$

**Corollary 1.4.** *Let  $e \in A$  be an idempotent. Let  $M$  be a left  $A$ -submodule of  $Ae$ . The following are equivalent.*

- (1)  $M$  is a direct summand of  $Ae$ .
- (2)  $M = Af$  for some idempotent  $f \leq e$  in  $A$ .

*Proof.* (2) $\Rightarrow$ (1):  $Ae = Af \oplus Af'$  where  $f' = e - f$  is an idempotent.

(1) $\Rightarrow$ (2): Let  $Ae = M \oplus N$ . Let  $\varphi : Ae \rightarrow Ae$  be the projection on  $M$  vanishing on  $N$ . Then  $\varphi \in \text{End}(Ae)$ . By Prop. 1.2,  $\varphi$  is the right multiplication by some  $f \in eAe$ . Since  $\varphi \circ \varphi = \varphi$ , clearly  $f^2 = f$ . Moreover,  $M = \varphi(Ae) = (Ae)f = Af$ .  $\square$

**Corollary 1.5.** *Let  $e \in A$  be an idempotent. The following are equivalent.*

- (1)  $Ae$  is an indecomposable left  $A$ -module.
- (2)  $e$  is primitive.

*Proof.* This follows immediately from Cor. 1.4.  $\square$

**Lemma 1.6.** *Let  $M$  be a nonzero finitely-generated left  $A$ -module. Then  $M$  has a maximal proper left  $A$ -submodule  $N$ . Consequently, there is an epimorphism of  $M$  onto an irreducible module.*

*Proof.* Let  $\xi_1, \dots, \xi_n$  generate  $M$ . Without loss of generality, we assume that  $\xi_1$  does not belong to the submodule  $N_0$  generated by  $\xi_2, \dots, \xi_n$ . By Zorn's lemma, there is a left submodule  $N \leq M$  maximal with respect to the property that  $N_0 \subset N$  and  $\xi_1 \notin N$ . Let us prove that  $N$  is a maximal proper submodule. Let  $N < K \leq M$ . Then by the maximality of  $N$  we must have  $\xi_1 \in K$ . So  $\xi_1, \dots, \xi_n \in K$ , and hence  $K = M$ . So  $K$  is not proper.  $\square$

## 2 Almost unital algebras

In this section, we introduce the notion of almost unital algebras, which is weaker than being almost unital and finite-dimensional.

**Definition 2.1.** We say that an algebra  $A$  is **almost unital** if the following conditions are satisfied:

- (a) For each  $x \in A$ , there is an idempotent  $e \in A$  such that  $x = exe$ .
- (b) For any finitely many idempotents  $e_1, \dots, e_n \in A$  there exists an idempotent  $e \in A$  such that  $e_i \leq e$  for all  $1 \leq i \leq n$ .

Throughout this section, unless otherwise stated,  $A$  is assumed to be almost unital.

**Definition 2.2.** We say that a left  $A$ -module  $M$  is **quasicoherent** if one of the following equivalent conditions holds:

- (1) For each  $\xi \in M$  we have  $\xi \in A\xi$ .
- (2) For each  $\xi \in M$  there exists an idempotent  $e \in A$  such that  $\xi = e\xi$ .

(3)  $M$  is a quotient module of  $\bigoplus_{i \in I} Ae_i$  where each  $e_i \in A$  is an idempotent.

(4)  $M$  is a quotient module of a free left  $A$ -module  $A^{\oplus I}$ .

The category of quasicohherent left  $A$ -modules is denoted by  $\mathbf{QCoh}_L(A)$ .

*Proof of equivalence.* (1) $\Rightarrow$ (2): For each  $\xi \in M$ , since  $\xi \in A\xi$ , we have  $\xi = a\xi$  for some  $a \in A$ . Choose idempotent  $e \in A$  such that  $a \in eAe$ . Then  $e\xi = ea\xi = a\xi = \xi$ .

(2) $\Rightarrow$ (1): Obvious.

(2) $\Rightarrow$ (3): For each  $\xi \in M$ , let  $e_\xi \in A$  be an idempotent such that  $e_\xi\xi = \xi$ . Then we have a morphism  $\bigoplus_{\xi \in M} Ae_\xi \rightarrow M$  whose restriction to  $Ae_\xi$  sends each  $a \in Ae_\xi$  to  $a\xi$ . Then  $\xi = e_\xi\xi$  implies that  $\xi \in Ae_\xi\xi$ , and hence  $\xi$  is in the range of this morphism. So this morphism is surjective.

(3) $\Rightarrow$ (4): This is obvious, since we have an epimorphism  $A \rightarrow Ae_i$  and hence an epimorphism  $\bigoplus_{i \in I} A \rightarrow \bigoplus_{i \in I} Ae_i$ .

(4) $\Rightarrow$ (2): It suffices to show that  $A^{\oplus I}$  satisfies the requirement of (2). Choose  $\xi = (a_i)_{i \in I} \in A^{\oplus I}$ . Then there are only finitely many  $i \in I$  such that  $a_i \neq 0$ . Since  $A$  is almost unital, there exist idempotents  $e_i \in A$  (where  $i \in I$ ) such that  $a_i = e_i a_i e_i$  for all  $i \in I$ . (If  $a_i = 0$ , then we choose  $e_i = 0$ ). Choose an idempotent  $e \in A$  such that  $e_i \leq e$  for all  $i \in I$ . Then  $\xi = e\xi$ .  $\square$

**Definition 2.3.** A left  $A$ -module  $M$  is called **coherent** if it is quasicohherent and finitely-generated. By the above proof of equivalence, it is clear that  $M$  is coherent iff  $M$  is a quotient of  $\bigoplus_{i \in I} Ae_i$  where  $I$  is a *finite* index set and  $e_i \in A$  is an idempotent. The category of coherent left  $A$ -modules is denoted by  $\mathbf{Coh}_L(A)$ .

However, note that a coherent left  $A$ -module is not necessarily a quotient of  $A^{\oplus n}$  where  $n \in \mathbb{Z}_+$ . Indeed,  $A$  is not necessarily finitely generated as a left  $A$ -module.

**Remark 2.4.** If  $M \in \mathbf{QCoh}_L(A)$ , then every submodule of  $M$  is quasicohherent, and every quotient module of  $M$  is quasicohherent. However, if  $M \in \mathbf{Coh}_L(A)$ , then a submodule of  $M$  is not known to be coherent. Thus,  $\mathbf{QCoh}_L(A)$  is an abelian category, while  $\mathbf{Coh}_L(A)$  is not known to be abelian.

**Proposition 2.5.** Let  $M \in \mathbf{QCoh}_L(A)$ . The following are equivalent.

(1)  $M$  is projective in the category of left  $A$ -modules.

(2)  $M$  is projective in  $\mathbf{QCoh}_L(A)$ .

(3)  $M$  is a direct summand of  $\bigoplus_{i \in I} Ae_i$  for some index set  $I$  and each  $e_i \in A$  is an idempotent.

*Proof.* (3) $\Rightarrow$ (1): It is well-known that a direct summand of a projective module is projective. Thus, it suffices to prove that  $\bigoplus_{i \in I} Ae_i$  is projective. Let  $\Phi : \bigoplus_{i \in I} Ae_i \rightarrow N$  be a morphism where  $N$  is a left  $A$ -module. Let  $\Gamma : K \rightarrow N$  be an epimorphism. Let

$$\eta_i = \Phi(e_i)$$

Since  $\Gamma$  is surjective, there is  $\xi_i \in K$  such that  $\Gamma(\xi_i) = \eta_i$ . Define  $\Psi : \bigoplus_{i \in I} Ae_i \rightarrow K$  to be the morphism sending each  $ae_i \in Ae_i$  to  $ae_i \xi_i$ . Then the following commute:

$$\begin{array}{ccc} & \bigoplus_{i \in I} Ae_i & \\ \Psi \swarrow & \downarrow \Phi & \searrow ae_i \\ K & \xrightarrow{\Gamma} N & \end{array} \quad \begin{array}{ccc} & ae_i & \\ \swarrow & \downarrow & \searrow \\ ae_i \xi_i & \xrightarrow{\quad} & ae_i \eta_i \end{array}$$

Note that  $\mapsto$  holds since  $\Gamma(ae_i \xi_i) = ae_i \Gamma(\xi_i) = ae_i \eta_i$ , and  $\downarrow$  holds since  $\Phi(ae_i) = \Phi(ae_i e_i) = ae_i \Phi(e_i) = ae_i \eta_i$ .

(1) $\Rightarrow$ (2): Obvious.

(2) $\Rightarrow$ (3): Choose an epimorphism  $\bigoplus_{i \in I} Ae_i \rightarrow M$ , which splits because  $M$  is projective. So  $M$  is a direct summand of  $\bigoplus_{i \in I} Ae_i$ .  $\square$

**Proposition 2.6.** *Let  $M \in \text{Coh}_L(A)$ . The following are equivalent.*

- (1)  $M$  is projective in the category of left  $A$ -modules.
- (2)  $M$  is projective in  $\text{QCoh}_L(A)$ .
- (3)  $M$  is projective in  $\text{Coh}_L(A)$ .
- (4)  $M$  is a direct summand of  $\bigoplus_{i \in I} Ae_i$  for some finite index set  $I$  and each  $e_i \in A$  is an idempotent.

Therefore, there is no ambiguity when talking about projective coherent left  $A$ -modules.

*Proof.* Clearly we have (1) $\Rightarrow$ (2) and (2) $\Rightarrow$ (3). By Prop. 2.5 we have (4) $\Rightarrow$ (1). Assume (3). By Def. 2.3, there is an epimorphism  $\bigoplus_{i \in I} Ae_i \rightarrow M$  such that  $I$  is finite, and that it splits (because  $M$  is projective in  $\text{Coh}_L(A)$ ). So (4) is true.  $\square$

**Remark 2.7.** If  $M \in \text{QCoh}_L(A)$ , clearly  $M$  is irreducible in  $\text{QCoh}_L(A)$  iff  $M$  is irreducible in the category of left  $A$ -modules; in this case we say that  $M$  is **irreducible**. Note that even if  $M \in \text{Coh}_L(A)$ , its irreducibility is understood as in  $\text{QCoh}_L(A)$  but not as in  $\text{Coh}_L(A)$ .

**Proposition 2.8.** *Let  $M$  be a left  $A$ -module. The following are equivalent.*

- (1)  $M \in \text{QCoh}_L(A)$  and  $M$  is irreducible.
- (2)  $M \simeq Ae/N$  where  $e \in A$  is an idempotent and  $N$  is a maximal proper left  $A$ -submodule of  $Ae$ .
- (3)  $M \simeq A/N$  where  $N$  is a maximal (proper) left ideal of  $A$ .

*Proof.* (1) $\Rightarrow$ (2): Let  $M \in \text{QCoh}_L(A)$  be irreducible. By Def. 2.2,  $M$  has an epimorphism  $\Phi$  from some  $\bigoplus_i Ae_i$  where  $e_i \in A$  is an idempotent. The restriction of  $\Phi$  to some  $Ae_i$  must be nonzero, and hence must be surjective (since  $M$  is irreducible). It follows that  $M$  has an epimorphism  $\Psi$  from  $Ae_i$ . Then  $N = \text{Ker} \Psi$  is a maximal proper left  $A$ -submodule of  $Ae_i$ , and  $M \simeq Ae_i/N$ .

(1) $\Rightarrow$ (3): In the above proof,  $M$  also has an epimorphism from  $\bigoplus_i A$  (since we have an epimorphism  $A \rightarrow Ae_i$ ). Thus, replacing  $Ae_i$  with  $A$  in the above proof, we are done.

(2),(3) $\Rightarrow$ (1): Clearly  $M$  is irreducible. That  $M \in \text{QCoh}_L(A)$  follows from Def. 2.2.  $\square$



### 3 Projective covers

Let  $A$  be an algebra, not necessarily almost unital. In this section, we recall some basic facts about projective covers. When  $A$  is unital, these results can be found in [AF92], for example. In the non-unital case, one can reduce to the unital setting by considering the unitalization of  $A$ . For the reader's convenience, we include complete proofs.

#### 3.1 Basic facts

**Definition 3.1.** Let  $M$  be a left  $A$ -module. A left  $A$ -submodule  $K \leq M$  is called **superfluous**, if for any left  $A$ -submodule  $L \leq M$  satisfying  $K + L = M$  we must have  $L = M$ .

**Remark 3.2.** Obviously, we have an equivalent description of superfluous submodules: Let  $\pi : M \rightarrow M/K$  be the quotient map. Then  $K \leq M$  is superfluous iff for any morphism of left  $A$ -modules  $\varphi : N \rightarrow M$  such that  $\pi \circ \varphi : N \rightarrow M/K$  is surjective, it must be true that  $\varphi$  is surjective.

**Definition 3.3.** Let  $M$  be a left  $A$ -module. A **projective cover** of  $M$  denotes a left  $A$ -module epimorphism  $\varphi : P \twoheadrightarrow M$  where  $P$  is a projective left  $A$ -module, and  $\text{Ker} \varphi$  is superfluous in  $P$ .

The following property says that among the projective modules that have epimorphisms to  $M$ , the projective cover is the smallest one in the sense of direct summand.

**Proposition 3.4.** Let  $\varphi : P \rightarrow M$  be a projective cover of  $M$ . Let  $\psi : Q \rightarrow M$  be an epimorphism where  $Q$  is projective. Then there is a morphism  $\alpha : Q \rightarrow P$  such that the following diagram commutes.

$$\begin{array}{ccc} & P & \\ \alpha \nearrow & \downarrow \varphi & \\ Q & \xrightarrow{\psi} & M \end{array} \quad (3.1)$$

Moreover, for any such  $\alpha$ , there is a left  $A$ -submodule  $P' \leq Q$  such that  $Q = \ker \alpha \oplus P'$  and that  $\alpha|_{P'} : P' \xrightarrow{\sim} P$  is an isomorphism.

By setting  $L = \ker \alpha$ , it follows that (3.1) is equivalent to

$$\begin{array}{ccc} & P & \\ 0 \oplus \text{id}_P \nearrow & \downarrow \varphi & \\ L \oplus P & \xrightarrow{0 \oplus \varphi} & M \end{array} \quad (3.2)$$

*Proof.* The existence of  $\alpha$  follows from the fact that  $Q$  is projective and that  $\varphi$  is an epimorphism. Moreover, since  $\ker \varphi$  is superfluous and  $\varphi \circ \alpha$  is surjective, by Rem. 3.2,  $\alpha$  is surjective. Therefore, since  $P$  is projective, the epimorphism  $\alpha$  splits, i.e., there is a morphism  $\beta : P \rightarrow Q$  such that  $\alpha \circ \beta : P \rightarrow P$  equals  $\text{id}_P$ . One sees that  $P' = \beta(P)$  fulfills the requirement.  $\square$

It follows that projective covers are unique up to isomorphisms:

**Corollary 3.5.** *Let  $M$  be a left  $A$ -module with projective covers  $\varphi : P \rightarrow M$  and  $\psi : Q \rightarrow M$ . Then there exists an isomorphism  $\alpha : Q \rightarrow P$  such that (3.1) commutes.*

*Proof.* By Prop. 3.4, there exists  $\alpha$  such that (3.1) commutes. It remains to show that  $\alpha$  is an isomorphism. We assume that (3.1) equals (3.2). Since  $0 \oplus \varphi : L \oplus P \rightarrow M$  is a projective cover,  $L + \ker(\varphi) = \ker(0 \oplus \varphi)$  is superfluous, and hence  $L$  is superfluous. Thus, since  $L + P$  equals  $Q = L \oplus P$ , we must have  $Q = P$  and hence  $L = 0$ . So  $\alpha = 0 \oplus \text{id}_P$  is an isomorphism.  $\square$

### 3.2 Projective covers of irreducibles

**Proposition 3.6.** *Suppose that  $\varphi : P \rightarrow M$  is a projective cover of an irreducible left  $A$ -module  $M$ . Then  $P$  is indecomposable.*

*Proof.* Suppose that  $P = P' \oplus P''$ . Then one of  $\varphi|_{P'}, \varphi|_{P''}$  (say  $\varphi|_{P'}$ ) is nonzero. Since  $M$  is irreducible,  $\varphi|_{P'} : P' \rightarrow M$  must be surjective. So the map  $P' \hookrightarrow P' \oplus P'' \xrightarrow{\varphi} M$  is surjective. Since  $\ker \varphi$  is superfluous, by Rem. 3.2,  $P' \hookrightarrow P' \oplus P''$  is surjective, and hence  $P'' = 0$ .  $\square$

**Theorem 3.7.** *Let  $e \in A$  be a primitive idempotent satisfying*

$$\dim eAe < +\infty$$

*Let  $K$  be any proper left  $A$ -submodule of  $Ae$ . Then  $K$  is superfluous. In other words, the quotient map  $Ae \rightarrow Ae/K$  is the projective cover of  $Ae/K$ .*

*Proof.* Step 1. Let  $\varphi : Ae \rightarrow Ae/K$  be the quotient map. Let  $N$  be a submodule of  $Ae$ . Assume that  $N + K = Ae$ ; in other words, if we let  $\iota : N \hookrightarrow Ae$  be the inclusion, then  $\varphi \circ \iota : N \rightarrow Ae/K$  is surjective. Our goal is to show that  $N = Ae$ .

Since  $Ae$  is projective and  $\varphi \circ \iota$  is surjective, there is a morphism  $\alpha : Ae \rightarrow N$  such that  $\varphi = \varphi \circ \iota \circ \alpha$ . Let  $\beta = \iota \circ \alpha$ . Then the following diagram commutes:

$$\begin{array}{ccccc} & & Ae & & \\ & \nearrow \alpha & \downarrow \varphi & \searrow \beta & \\ N & \xleftarrow{\iota} & Ae & \xrightarrow{\varphi} & Ae/K \end{array} \quad (3.3)$$

To prove that  $\iota$  is surjective, it suffices to show that  $\beta$  is surjective.

Step 2. Suppose that  $\beta$  is not surjective. Let us find a contradiction. Since  $\beta \in \text{End}_{A,-}(Ae)$ , by Prop. 1.2,  $\beta$  is the right multiplication by some  $x \in eAe$ . Let  $R_x : eAe \rightarrow eAe$  be the right multiplication of  $x$  on  $eAe$ . Then  $R_x$  is not surjective. Otherwise, there exists  $a \in A$  such that  $R_x(eae) = e$ , i.e.,  $eaex = e$ . Then for each  $b \in A$ , we have  $be = beaex = \beta(beae)$ , contradicting the fact that  $\beta$  is not surjective.

It is well-known that if  $T$  is a linear operator on a finite-dimensional  $\mathbb{C}$ -vector space  $W$ , then  $W$  is the direct sum of generalized eigenspaces of  $T$ , and the projection operator of  $W$  onto each generalized eigenspace is a polynomial of  $T$ . Therefore,  $R_x$  has only one

eigenvalue. Otherwise, there is a polynomial  $p$  such that  $p(R_x) = R_{p(x)}$  is the projection of  $eAe$  onto a proper subspace, and hence  $p(x)$  is an idempotent in  $eAe$  not equal to 0 or  $e$ . This is impossible, since  $e$  is assumed to be primitive.

Therefore,  $R_x$  has a unique eigenvalue, which must be 0 since  $R_x$  is not surjective. By linear algebra,  $R_x$  is nilpotent. Since  $R_{x^n} = (R_x)^n$ , it follows that  $x$  is nilpotent, and hence  $\beta$  is nilpotent. By (3.3), we have  $\varphi = \varphi \circ \beta$ , and hence  $\varphi = \varphi \circ \beta = \varphi \circ \beta^2 = \varphi \circ \beta^3 = \dots = 0$ . This contradicts the fact that  $\varphi$  is a surjection onto a nonzero module, finishing the proof.  $\square$

**Corollary 3.8.** *Let  $e \in A$  be a primitive idempotent satisfying  $\dim eAe < +\infty$ . Then  $Ae$  has a unique proper maximal left  $A$ -submodule, denoted by  $\text{rad}(Ae)$ .*

It follows from Thm. 3.7 that  $Ae$  is the projective cover of the irreducible  $Ae/\text{rad}(Ae)$ .

*Proof.* By Lem. 1.6,  $Ae$  has at least one proper maximal left  $A$ -submodule. Suppose that  $K \neq L$  are proper maximal left  $A$ -submodules of  $Ae$ . By the maximality, we have  $K + L = Ae$ . By Thm. 3.7,  $L$  is superfluous. So  $K = Ae$ , impossible.  $\square$

## 4 Left pseudotraces

Let  $A, B$  be algebras such that  $B$  is unital. Fix an  $A$ - $B$  bimodule  $M$ . We do not assume that  $M_B$  is unital, i.e.,  $1_B \in B$  acts as the identity on  $M$ .

**Definition 4.1.** A **left coordinate system** of  $M$  denotes a collection of morphisms

$$\alpha_i \in \text{Hom}_{-,B}(B, M) \quad \check{\alpha}^i \in \text{Hom}_{-,B}(M, B) \quad (4.1)$$

where  $i$  runs through an index set  $I$  such that the following conditions hold:

- (a) For each  $\xi \in M$ , we have  $\check{\alpha}^i(\xi) = 0$  for all but finitely many  $i \in I$ , and  $\sum_{i \in I} \alpha_i \circ \check{\alpha}^i(\xi) = \xi$ .
- (b) For each  $x \in A$  (viewed as an element of  $\text{End}_{-,B}(M)$ ), we have  $x \circ \alpha_i = 0$  and  $\check{\alpha}^i \circ x = 0$  for all but finitely many  $i \in I$ .

**Remark 4.2.**  $M$  is a projective right  $B$ -module iff there exists  $(\alpha_i, \check{\alpha}^i)_{i \in I}$  of the form (4.1) satisfying condition (a).

*Proof.* Suppose that there exists  $(\alpha_i, \check{\alpha}^i)_{i \in I}$  such that (a) holds. Define morphisms of right  $B$ -modules

$$\begin{aligned} \Phi : B^{\oplus I} &\rightarrow M & \oplus_i b_i &\mapsto \sum_i \alpha_i(b_i) \\ \Psi : M &\rightarrow B^{\oplus I} & \xi &\mapsto \oplus_i \check{\alpha}^i(\xi) \end{aligned}$$

Then (a) implies that  $\Phi \circ \Psi = \text{id}_M$ . Thus,  $M$  is a direct summand of  $B^{\oplus I}$ , and hence is projective as a right  $B$ -module.

Conversely, assume  $M$  is projective as a right  $B$ -module. Then we have an epimorphism  $\Phi : B^{\oplus I} \rightarrow M$  and a morphism  $\Psi : M \rightarrow B^{\oplus I}$  such that  $\Phi \circ \Psi = \text{id}_M$ . For each

$i \in I$ , let  $\iota_i : B \rightarrow B^{\oplus I}$  be the inclusion map of  $B$  into the  $i$ -th direct summand, and  $\pi_i : B^{\oplus I} \rightarrow B$  be the projection map onto the  $i$ -th direct summand. Set

$$\alpha_i = \Phi \circ \iota_i \quad \check{\alpha}^i = \pi_i \circ \Psi$$

Then  $(\alpha_i, \check{\alpha}^i)_{i \in I}$  satisfies (a). □

**Definition 4.3.** Assume that  $M$  has a left coordinate system  $(\alpha_i, \check{\alpha}^i)_{i \in I}$ . Define the  **$B$ -trace function**

$$\mathrm{Tr}^B : A \rightarrow B/[B, B] \quad x \mapsto \sum_{i \in I} \check{\alpha}^i \circ x \circ \alpha_i$$

where the RHS, originally an element of  $\mathrm{End}_{-,B}(B) \simeq B$ <sup>2</sup> is descended to  $B/[B, B]$ .

**Lemma 4.4.** The definition of  $\mathrm{Tr}^B$  is independent of the choice of left coordinate systems.

*Proof.* Suppose that  $(\beta_j, \check{\beta}^j)_{j \in J}$  is another left coordinate system of the  $A$ - $B$  bimodule  $M$ . Let  $I_x \subset I$  and  $J_x \subset J$  be finite sets such that  $\check{\alpha}^i \circ x = 0, x \circ \alpha_i = 0$  for any  $i \in I \setminus I_x$ , and that  $\check{\beta}^j \circ x = 0, x \circ \beta_j = 0$  for any  $j \in J \setminus J_x$ . Then

$$\sum_{i \in I_x} \check{\alpha}^i \circ x \circ \alpha_i = \sum_{i \in I_x, j \in J} \check{\alpha}^i \circ x \circ \beta_j \circ \check{\beta}^j \circ \alpha_i = \sum_{i \in I_x, j \in J_x} \check{\alpha}^i \circ x \circ \beta_j \circ \check{\beta}^j \circ \alpha_i$$

Since each  $\check{\alpha}^i \circ x \circ \beta_j$  and  $\check{\beta}^j \circ \alpha_i$  are in  $\mathrm{End}_{-,B}(B) \simeq B$ , the RHS above equals

$$\sum_{i \in I_x, j \in J_x} \check{\beta}^j \circ \alpha_i \circ \check{\alpha}^i \circ x \circ \beta_j = \sum_{j \in J_x} \check{\beta}^j \circ x \circ \beta_j$$

in  $B/[B, B]$ . □

**Proposition 4.5.**  $\mathrm{Tr}^B$  is **symmetric**, i.e.,  $\mathrm{Tr}^B(xy) = \mathrm{Tr}^B(yx)$  for any  $x, y \in A$ . Therefore,  $\mathrm{Tr}^B$  descends to a linear map  $A/[A, A] \rightarrow B/[B, B]$ .

*Proof.* Let  $x, y \in A$ . Let  $I_0 \subset I$  be a finite set such that  $\check{\alpha}^i \circ x = \check{\alpha}^i \circ y = 0$  and  $x \circ \alpha_i = y \circ \alpha_i = 0$  for all  $i \in I \setminus I_0$ . Then

$$\mathrm{Tr}^B(xy) = \sum_{i \in I_0} \check{\alpha}^i \circ x \circ y \circ \alpha_i = \sum_{i, j \in I_0} \check{\alpha}^i \circ x \circ \alpha_j \circ \check{\alpha}^j \circ y \circ \alpha_i$$

and similarly

$$\mathrm{Tr}^B(yx) = \sum_{i, j \in I_0} \check{\alpha}^j \circ y \circ \alpha_i \circ \check{\alpha}^i \circ x \circ \alpha_j$$

The two RHS's are equal in  $B/[B, B]$ , noting that  $\check{\alpha}^i \circ x \circ \alpha_j$  and  $\check{\alpha}^j \circ y \circ \alpha_i$  are both in  $\mathrm{End}_{-,B}(B) \simeq B$ . □

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<sup>2</sup>This isomorphism relies on the fact that  $B$  is unital.

**Definition 4.6.** Let  $\phi : B \rightarrow \mathbb{C}$  be a **symmetric linear functional (SLF)**, i.e., a linear map satisfying  $\phi(ab) = \phi(ba)$  for all  $a, b \in B$ . The (left) **pseudotraces** associated to  $\phi$  (and  $M$ ), denoted by  $\text{Tr}^\phi$ , is defined to be

$$\text{Tr}^\phi = \phi \circ \text{Tr}^B : A \rightarrow \mathbb{C} \quad (4.2)$$

It is an SLF on  $A$ .

Thus, for each  $x \in A$  we have

$$\text{Tr}^\phi(x) = \sum_{i \in I} \phi(\tilde{\alpha}^i \circ x \circ \alpha_i(1_B)) \quad (4.3)$$

## 5 AUF algebras and projective covers of irreducibles

**Definition 5.1.** An algebra  $A$  is called **almost unital and finite-dimensional (AUF)** if there is a family of mutually orthogonal idempotents  $(e_i)_{i \in \mathfrak{I}}$  such that the following conditions hold:

- (a) For each  $i, j \in \mathfrak{I}$  we have  $\dim e_i A e_j < +\infty$ .
- (b)  $A = \sum_{i, j \in \mathfrak{I}} e_i A e_j$ . (That is, for each  $x \in A$  one can find a finite subset  $I \subset \mathfrak{I}$  and a collection  $(x_{i,j})_{i, j \in \mathfrak{I}}$  such that  $x = \sum_{i, j \in I} e_i x_{i,j} e_j$ .)

Note that (b) automatically implies  $A = \bigoplus_{i, j \in \mathfrak{I}} e_i A e_j$ .

It is illuminating to view an element  $x \in A$  as an  $\mathfrak{I} \times \mathfrak{I}$  matrix whose  $(i, j)$ -entry is  $e_i x e_j$ .

**Remark 5.2.** Each AUF algebra  $A$  is almost unital.

*Proof.* For each  $x_1, \dots, x_n \in A$ , we can find a subset  $I_0 \subset \mathfrak{I}$  such that  $x_1, \dots, x_n \in e' A e'$ , where  $e' = \sum_{i \in I_0} e_i$ . By choosing  $n = 1$  and  $x_1 = x \in A$ , we see  $x = e' x e'$ . By choosing idempotents  $x_i = e_i \in A$ , we see  $e_i \leq e'$  for all  $1 \leq i \leq n$ .  $\square$

**Lemma 5.3.** In Def. 5.1, one can assume moreover that each  $e_i$  is primitive (in  $A$ ).

*Proof.* Let  $(e_i)_{i \in \mathfrak{I}}$  be as in Def. 5.1. For each  $i \in \mathfrak{I}$ , since  $e_i A e_i$  is a finite-dimensional left  $e_i A e_i$ -module, it is a finite direct sum of indecomposable left  $e_i A e_i$ -submodules. By Cor. 1.4 and 1.5, we have a finite direct sum  $e_i A e_i = \bigoplus_{k \in \mathfrak{K}_i} e_i A f_{i,k}$  where  $(f_{i,k})_{k \in \mathfrak{K}_i}$  is a finite family of mutually orthogonal idempotents in  $e_i A e_i$ , that  $\sum_k f_{i,k} = e_i$ , and that each  $f_{i,k}$  is primitive in  $e_i A e_i$ . Clearly  $f_{i,k}$  is also primitive in  $A$ . Replacing  $(e_i)_{i \in \mathfrak{I}}$  by  $(f_{i,k})_{i \in \mathfrak{I}, k \in \mathfrak{K}_i}$  does the job.  $\square$

In the remaining part of this section, we always assume that  $A$  is AUF.

**Remark 5.4.** For any idempotents  $e, f \in A$ , we have

$$\dim e A f < +\infty$$

Indeed, one can find a finite set  $I_0 \subset \mathcal{I}$  such that  $e, f \in e' Ae'$  where  $e' = \sum_{i \in I_0} e_i$ . Then  $\dim e' Ae' < +\infty$ , and hence  $\dim eAf < +\infty$ .

It follows that each idempotent  $e \in A$  has a (finite) orthogonal primitive decomposition  $e = \varepsilon_1 + \cdots + \varepsilon_n$ . This follows from a decomposition of the finite-dimensional left  $eAe$ -module  $eAe$  into indecomposable submodules.  $\square$

Recall Rem. 2.7 about irreducibility.

**Theorem 5.5.** *The following are true.*

1. For each primitive idempotent  $e \in A$ , let  $\text{rad}(Ae)$  be the unique proper maximal left submodule of  $Ae$  (cf. Cor. 3.8). Then  $Ae \rightarrow Ae/\text{rad}(Ae)$  gives a projective cover of the irreducible coherent module  $Ae/\text{rad}(Ae)$ .
2. Any irreducible  $M \in \text{QCoh}_L(A)$  is isomorphic to  $Ae/\text{rad}(Ae)$  for some primitive idempotent  $e \in A$ .
3. Let  $e, f$  be primitive idempotents. Then the following are equivalent:
  - (1)  $Ae \simeq Af$  as left  $A$ -modules.
  - (2)  $Ae/\text{rad}(Ae) \simeq Af/\text{rad}(Af)$  as left  $A$ -modules.
  - (3)  $e \simeq f$ , i.e., there is a partial isometry (in  $A$ ) from  $e$  to  $f$ .

*Proof.* Part 1 was already proved, cf. Thm. 3.7. (Note that Thm. 3.7 and its consequences are applicable since  $\dim eAe < +\infty$  by Rem. 5.4.)

Part 2: By Prop. 2.8,  $M$  has an epimorphism  $\Psi$  from  $A$ . Let  $(e_i)_{i \in \mathcal{I}}$  be as in Def. 5.1 such that each  $e_i$  is primitive (Lem. 5.3). Then  $A \simeq \bigoplus_i Ae_i$  as left  $A$ -modules. The restriction of  $\Psi$  to some  $Ae_i$  must be nonzero, and hence must be surjective. Therefore  $M \simeq Ae_i/\text{rad}(Ae_i)$ .

Part 3: (1) $\Rightarrow$ (2) is obvious. (2) $\Rightarrow$ (1) follows from the uniqueness of projective covers (Cor. 3.5). (1) $\Leftrightarrow$ (3) follows from Cor. 1.3.  $\square$

**Corollary 5.6.** *Let  $P \in \text{Coh}_L(A)$ . The following are equivalent.*

- (1)  $P$  is projective and indecomposable.
- (2)  $P$  is the projective cover of an irreducible  $M \in \text{QCoh}_L(A)$ , and hence (by Thm. 5.5)  $P$  is isomorphic to  $Ae$  for some primitive idempotent  $e \in A$ .

*Proof.* (2) $\Rightarrow$ (1): This follows from Prop. 3.6.

(1) $\Rightarrow$ (2): By Lem. 1.6,  $P$  has an epimorphism to an irreducible, which (by Thm. 5.5) is of the form  $Ae/\text{rad}(Ae)$  where  $e \in A$  is a primitive idempotent. We know that  $Ae$  is its projective cover. Since  $P$  is projective, by Prop. 3.4,  $Ae$  is a direct summand of  $P$ . Since  $P$  is indecomposable, we must have  $P \simeq Ae$ .  $\square$

## 6 Pseudotraces and generating idempotents of strongly AUF algebras

Let  $A$  be AUF. In this section, we show that if  $e \in A$  is a generating idempotent, any SLF  $\psi$  on  $A$  can be recovered from  $\psi|_{eAe}$  via the pseudotrace construction.

**Definition 6.1.** An idempotent  $e \in A$  is called **generating** if every irreducible  $M \in \text{QCoh}_L(A)$  has an epimorphism from  $Ae$ .

**Proposition 6.2.** Let  $e \in A$  be an idempotent. Let  $e = \varepsilon_1 + \cdots + \varepsilon_n$  be an orthogonal primitive decomposition (cf. Rem. 5.4). The following are equivalent:

- (1)  $e$  is generating.
- (2) Any primitive idempotent of  $A$  is equivalent to  $\varepsilon_i$  for some  $i$ .
- (3) Any irreducible  $M \in \text{QCoh}_L(A)$  is isomorphic to  $A\varepsilon_i/\text{rad}(A\varepsilon_i)$  for some  $i$ .

*Proof.* (1) $\Rightarrow$ (3): Each irreducible  $M \in \text{QCoh}_L(A)$  has an epimorphism from  $Ae = A\varepsilon_1 \oplus \cdots \oplus A\varepsilon_n$ , and hence an epimorphism from some  $A\varepsilon_i$ . By Cor. 3.8, the kernel of this epimorphism is  $\text{rad}(A\varepsilon_i)$ . Therefore, we have  $A\varepsilon_i/\text{rad}(A\varepsilon_i) \simeq M$ .

(3) $\Rightarrow$ (1): Obvious.

(2) $\Leftrightarrow$ (3): Immediate from Thm. 5.5. □

**Corollary 6.3.** Let  $e, f \in A$  be idempotents such that  $e$  is a generating idempotent of  $A$  and  $e \leq f$ . Then  $e$  is a generating idempotent of  $fAf$ .

*Proof.* Let  $p$  be any primitive idempotent of  $fAf$ . Then  $p$  is a primitive idempotent of  $A$ . By Prop. 6.2, if we let  $e = \varepsilon_1 + \cdots + \varepsilon_n$  be an orthogonal primitive decomposition, then there exist  $1 \leq i \leq n$  and  $u \in \varepsilon_i Ap, v \in pA\varepsilon_i$  such that  $uv = \varepsilon_i$  and  $vu = p$ . So  $p$  is equivalent in  $fAf$  to  $\varepsilon_i$ . By Prop. 6.2, we conclude that  $e$  is generating in  $fAf$ . □

**Corollary 6.4.** The following are equivalent.

- (1)  $A$  has a generating idempotent.
- (2)  $\text{QCoh}_L(A)$  has finitely many isomorphism classes of irreducible objects.
- (3)  $A$  has finitely many equivalence classes of primitive idempotents.

If one of these conditions holds, we say that  $A$  is **strongly AUF**.

*Proof.* (1) $\Rightarrow$ (2): Immediate from Prop. 6.2.

(2) $\Leftrightarrow$ (3): Immediate from Thm. 5.5.

(3) $\Rightarrow$ (1): Let  $M_1, \dots, M_n \in \text{QCoh}_L(A)$  exhaust all isomorphism classes of irreducibles. Let  $(e_i)_{i \in \mathcal{J}}$  be as in Def. 5.1. For each  $1 \leq k \leq n$ , by Prop. 2.8,  $M_k$  has an epimorphism from  $A$ . Since  $A = \bigoplus_{i \in \mathcal{J}} Ae_i$ , it follows that  $M_k$  has an epimorphism from  $Ae_{i_k}$  for some  $i_k \in \mathcal{J}$ . If we assume at the beginning that  $M_1, \dots, M_n$  are mutually non-isomorphic, then  $e_{i_1}, \dots, e_{i_n}$  must be distinct, and hence mutually orthogonal. So  $e = e_{i_1} + \cdots + e_{i_n}$  is a generating idempotent. □

**Theorem 6.5.** Assume that  $A$  is strongly AUF, and let  $e \in A$  be a generating idempotent. Then the  $A$ -( $eAe$ ) bimodule  $Ae$  has a left coordinate system. In particular, by Rem. 4.2,  $Ae$  is a projective right  $eAe$ -module.

The following construction of a left coordinate system is important and is motivated by [Ari10, Lem. 3.9].

*Proof.* Let  $(e_i)_{i \in \mathcal{I}}$  be as in Def. 5.1. By Lem. 5.3, we can assume that each  $e_i$  is primitive. Let  $e = \varepsilon_1 + \cdots + \varepsilon_n$  be an orthogonal primitive decomposition of  $e$ . By Prop. 6.2, there are partial isometries  $u_i, v_i$  such that

$$\begin{aligned} v_i u_i &= \varepsilon_{k_i} & u_i v_i &= e_i \\ u_i &\in e_i A \varepsilon_{k_i} & v_i &\in \varepsilon_{k_i} A e_i \end{aligned}$$

where  $k_i \in \{1, \dots, n\}$ . In particular  $u_i \in e_i A e$  and  $v_i \in e A e_i$ . Let

$$\begin{aligned} \alpha_i &\in \text{Hom}_{-, eAe}(eAe, Ae) & \check{\alpha}^i &\in \text{Hom}_{-, eAe}(Ae, eAe) \\ \alpha_i(xe) &= u_i \cdot x e & \check{\alpha}^i(xe) &= v_i \cdot x e \end{aligned}$$

One checks easily that  $(\alpha_i, \check{\alpha}^i)_{i \in \mathcal{I}}$  is a left coordinate system. □

The proof of [Ari10, Thm. 3.10] can be easily adapted to prove the following theorem.

**Theorem 6.6.** Assume that  $A$  is strongly AUF, and let  $e \in A$  be a generating idempotent. Then there is a linear isomorphism

$$\text{SLF}(A) \xrightarrow{\cong} \text{SLF}(eAe) \quad \psi \mapsto \psi|_{eAe}$$

whose inverse is given by

$$\text{SLF}(eAe) \xrightarrow{\cong} \text{SLF}(A) \quad \phi \mapsto \text{Tr}^\phi$$

Here,  $\text{Tr}^\phi$  is the pseudotrace on  $A$  with respect to  $\phi$  and the  $A$ -( $eAe$ ) bimodule  $Ae$ .

*Proof.* Let  $u_i, v_i, \alpha_i, \check{\alpha}^i$  be as in the proof of Thm. 6.5. For any  $\phi \in \text{SLF}(eAe)$ , let us compute  $\text{Tr}^\phi$ . Let  $x \in A$ , viewed as an element of  $\text{End}_{-, eAe}(Ae)$ . Then  $\check{\alpha}^i \circ x \circ \alpha_i \in \text{End}_{-, eAe}(eAe)$  equals (the left multiplication by)  $v_i x u_i$ . Then

$$\text{Tr}^\phi(x) = \sum_{i \in \mathcal{I}} \phi(v_i x u_i) \tag{6.1}$$

Note that the RHS is a finite sum since  $u_i = e_i u_i$ , and since  $x e_i = 0$  for all but finitely many  $i$ .

To show that  $\text{Tr}^\phi|_{eAe} = \phi$ , we compute

$$\text{Tr}^\phi(xe) = \sum_i \phi(v_i x e u_i) = \sum_i \phi(v_i x e \cdot e u_i)$$

Since  $v_i x e, e u_i \in eAe$ , and since  $\phi$  is SLF, we have

$$\text{Tr}^\phi(xe) = \sum_i \phi(e u_i \cdot v_i x e) = \sum_i \phi(e e_i x e) = \phi(xe)$$



Finally, let  $\psi \in \text{SLF}(A)$ . Then for each  $x \in A$ ,

$$\text{Tr}^{\psi|_{eAe}}(x) = \sum_i \psi|_{eAe}(v_i x u_i) = \sum_i \psi(v_i x u_i) = \sum_i \psi(u_i v_i x) = \sum_i \psi(e_i x) = \psi(x)$$

This proves  $\text{Tr}^{\psi|_{eAe}} = \psi$ . □

## 7 Projective generators of strongly AUF algebras

Let  $A$  be an AUF algebra.

**Remark 7.1.** A left  $A$ -module  $M$  is coherent if and only if  $M$  is a quotient module of  $(Ae)^{\oplus n}$  where  $n \in \mathbb{Z}_+$  and  $e \in A$  is an idempotent.

*Proof.* “ $\Leftarrow$ ” is obvious. Conversely, let  $M \in \text{Coh}_L(A)$ . By Def. 2.3,  $M$  is a quotient module of  $Ap_1 \oplus \cdots \oplus Ap_n$  where each  $p_i$  is an idempotent. By Rem. 5.2, one can find an idempotent  $e \in A$  which is  $\geq p_1, \dots, p_n$ . Then  $M$  is a quotient module of  $(Ae)^{\oplus n}$ . □

**Remark 7.2.** By Rem. 7.1, if  $M \in \text{Coh}_L(A)$  and  $x \in A$ , then  $\dim xM < +\infty$ .

*Proof.* Suppose that  $M$  has an epimorphism from  $N := (Ae)^{\oplus n}$  where  $e \in A$  is an idempotent. Then  $\dim xM \leq \dim xN$ . Let  $f \in A$  be an idempotent such that  $x = fxf$ . Then  $xAe \subset fAe$ , and hence

$$\dim xN = n \dim xAe \leq n \dim fAe < +\infty$$

□

### 7.1 Basic facts

**Definition 7.3.** Let  $\mathcal{S}$  and  $\mathcal{T}$  be classes of objects in  $\text{Coh}_L(A)$ . We say that  $\mathcal{S}$  **generates**  $\mathcal{T}$  if each object of  $\mathcal{T}$  is a quotient of a *finite* direct sum of objects in  $\mathcal{S}$ .

**Definition 7.4.** We say that  $M \in \text{Coh}_L(A)$  is a **generator** (of  $\text{Coh}_L(A)$ ) if it generates every object of  $\text{Coh}_L(A)$ , i.e., every  $N \in \text{Coh}_L(A)$  is a quotient module of  $M^{\oplus n}$  for some  $n \in \mathbb{Z}_+$ . A generator which is also projective is called a **projective generator**.

**Example 7.5.** Let  $(e_i)_{i \in \mathcal{I}}$  be as in Def. 5.1. Then  $\mathcal{S} := \{Ae_i : i \in \mathcal{I}\}$  generates  $\text{Coh}_L(A)$ .

*Proof.* By the proof of Rem. 5.2, for any idempotent  $e \in A$  one can find a finite set  $I_0 \subset \mathcal{I}$  such that  $e \leq \sum_{i \in I_0} e_i$ . Therefore,  $\mathcal{S}$  generates each  $Ae$ , and hence (by Rem. 7.1) generates  $\text{Coh}_L(A)$ . □

**Proposition 7.6.** Let  $M \in \text{Coh}_L(A)$  be projective. The following are equivalent.

- (1)  $M$  is a projective generator.
- (2) Each irreducible  $N \in \text{Coh}_L(A)$  has an epimorphism from  $M$ .

*Proof.* (1) $\Rightarrow$ (2): Obvious.

(2) $\Rightarrow$ (1): Let  $(e_i)_{i \in \mathcal{I}}$  be as in Def. 5.1. By Lem. 5.3, we assume that each  $e_i$  is primitive. By Exp. 7.5, it suffices to prove that  $M$  generates each  $Ae_i$ . By Thm. 5.5,  $Ae_i$  is the projective cover of the irreducible  $N := Ae_i/\text{rad}(Ae_i)$ . By (2),  $M$  has an epimorphism to  $N$ . Since  $M$  is projective, by Prop. 3.4,  $Ae_i$  is isomorphic to a direct summand of  $M$ .  $\square$

**Corollary 7.7.** *Let  $e \in A$  be an idempotent. Then the following are equivalent.*

- (1)  $Ae$  is a (necessarily projective) generator.
- (2)  $e$  is a generating idempotent.

*Proof.* (1) $\Rightarrow$ (2): Clear from Def. 6.1. (2) $\Rightarrow$ (1): Immediate from Prop. 7.6.  $\square$

**Proposition 7.8.**  *$\text{Coh}_L(A)$  has a projective generator if and only if  $A$  is strongly AUF.*

*Proof.* “ $\Leftarrow$ ” follows from Cor. 6.4 and 7.7. Conversely, if  $\text{Coh}_L(A)$  has a projective generator  $M$ , by Rem. 7.1, an idempotent  $e \in A$  can be found such that  $Ae$  generates  $M$ , and hence generates  $\text{Coh}_L(A)$ . So  $e$  is a generating idempotent. Thus, by Cor. 6.4,  $\text{Coh}_L(A)$  has finitely many irreducibles. So  $A$  is strongly AUF.  $\square$

## 7.2 Projective generators and endomorphism algebras

Our next goal is to give criteria for projective generators in terms of the endomorphism algebras. We need the endomorphism algebras to be finite-dimensional:

**Proposition 7.9.** *Let  $M, N \in \text{Coh}_L(A)$ . Then*

$$\dim \text{Hom}_{A,-}(M, N) < +\infty$$

*Proof.* By Def. 2.3, there is an epimorphism from a finite direct sum  $\bigoplus_i Ae_i$  to  $M$ , where  $e_i$  is an idempotent. By taking composition with this epimorphism, we get

$$\text{Hom}_{A,-}(M, N) \rightarrow \text{Hom}_{A,-}\left(\bigoplus_i Ae_i, N\right) \simeq \bigoplus_i \text{Hom}_{A,-}(Ae_i, N) \quad (7.1)$$

where the first map is injective. Thus, it suffices to prove that each  $\text{Hom}_{A,-}(Ae_i, N)$  is finite-dimensional.

Again, we can find an epimorphism  $\Phi : \bigoplus_j Af_j \twoheadrightarrow N$  (where  $\bigoplus_j$  is finite). Since  $Ae_i$  is projective, each  $\alpha \in \text{Hom}_{A,-}(Ae_i, N)$  can be lifted to some  $\beta \in \text{Hom}_{A,-}(Ae_i, \bigoplus_j Af_j)$  such that  $\alpha = \Phi \circ \beta$ . Thus

$$\dim \text{Hom}_{A,-}(Ae_i, N) \leq \dim \text{Hom}_{A,-}\left(Ae_i, \bigoplus_j Af_j\right) = \sum_j \dim \text{Hom}_{A,-}(Ae_i, Af_j)$$

where  $\dim \text{Hom}_{A,-}(Ae_i, Af_j) = \dim e_i Af_j < +\infty$ .  $\square$

**Proposition 7.10.** *Let  $M$  be a left  $A$ -module. Let  $B = \text{End}_{A,-}(M)^{\text{op}}$ , and let  $p, q \in B$  be idempotents. Then an element of  $\text{Hom}_{A,-}(Mp, Mq)$  is precisely the right multiplication of an element of  $pBq$ . In particular, we have a canonical isomorphism*

$$\text{End}_{A,-}(Mp)^{\text{op}} \simeq pBp$$

*Consequently, the direct summands of the left  $A$ -module  $Mp$  correspond bijectively to the sub-idempotents of  $p$  in  $B$ .*

*Proof.* This is similar to the proofs of Prop. 1.2 and Cor. 1.4. Any  $y \in pBq$  defines a morphism  $Mp \rightarrow Mq$  by right multiplication. Conversely, if  $T \in \text{Hom}_{A,-}(Mp, Mq)$ , let  $\hat{T} : M \rightarrow M$  be  $\hat{T}(\xi) = T(\xi p)$ . Then  $\hat{T} \in \text{End}_{A,-}(M)$ , and hence  $\hat{T}$  is the right multiplication by some  $\hat{y} \in B$ . Note that  $T = \hat{T}|_{Mp}$ , and hence  $T(\xi p) = \xi p \hat{y}$  for each  $\xi \in M$ . Since  $T$  has range in  $Mq$ , we have  $T(\xi p) = \xi p \hat{y} q$ . So  $T$  is the right multiplication by  $y := p \hat{y} q \in pBq$ .  $\square$

**Theorem 7.11.** *Let  $M \in \text{Coh}_L(A)$ . Let  $B = \text{End}_{A,-}(M)^{\text{op}}$  which is a finite-dimensional unital algebra (by Prop. 7.9). Let  $p \in B$  be an idempotent. Consider the following statements:*

- (1) *As coherent left  $A$ -modules,  $Mp$  generates  $M$ .*
- (2)  *$p$  is a generating idempotent of  $B$ .*

*Then (2) $\Rightarrow$ (1). If  $M$  is projective, then (1) $\Leftrightarrow$ (2).*

*Proof.* (2) $\Rightarrow$ (1): Since  $\dim B < +\infty$ , we have a primitive orthogonal decomposition  $1_B = q_1 + \cdots + q_n$  where each  $q_j \in B$  is a primitive idempotent. By Prop. 6.2, each  $q_j$  is equivalent to a sub-idempotent of  $p$ . Thus  $Mq_j$  is isomorphic to a direct summand of the left  $A$ -module  $Mp$ . So  $Mp$  generates  $\bigoplus_j Mq_j = M$ .

(1) $\Rightarrow$ (2): Let  $q$  be any primitive idempotent of  $B$ . Since  $Mp$  generates  $M$  and since  $M$  generates  $Mq$ , we have that  $Mp$  generates  $Mq$ . We claim that  $Mq$  is isomorphic to a direct summand of  $Mp$ . Then Prop. 7.10 will imply that  $q$  is equivalent (in  $B$ ) to a sub-idempotent of  $p$ . This implies (2), thanks to Prop. 6.2.

Let us prove the claim, assuming that  $M$  is projective. Since  $Mq$  is a direct summand of  $M$ , we see that  $Mq$  is projective. Since  $q$  is primitive in  $B$ , by Prop. 7.10,  $Mq$  is an indecomposable left  $A$ -module. Therefore, by Cor. 5.6,  $Mq$  is the projective cover of an irreducible  $N \in \text{Coh}_L(A)$ . Since  $Mp$  generates  $Mq$ , it generates  $N$ . Thus  $N$  has an epimorphism from a finite direct sum of  $Mp$ . Since  $N$  is irreducible,  $N$  has an epimorphism from  $Mp$ . Note that  $Mp$  is also projective. Therefore, by Prop. 3.4,  $Mq$  is isomorphic to a direct summand of  $Mp$ .  $\square$

**Corollary 7.12.** *Assume that  $G \in \text{Coh}_L(A)$  is a projective generator. Let  $M$  be a left  $A$ -module. Then the following are equivalent.*

- (1)  *$M \in \text{Coh}_L(A)$ , and  $M$  is a projective generator (of  $\text{Coh}_L(A)$ ).*
- (2) *There exist  $n \in \mathbb{Z}_+$  and a generating idempotent  $p$  of  $B := \text{End}_{A,-}(G^{\oplus n})^{\text{op}}$  such that  $M \simeq G^{\oplus n} \cdot p$ .*

In particular, if  $e \in A$  is a generating idempotent, one can take  $G = Ae$ . Thus a projective generator of  $\text{Coh}_L(A)$  is (up to isomorphisms) precisely of the form  $(Ae)^{\oplus n}p$  where  $n \in \mathbb{Z}_+$  and  $p \in \text{End}_{A,-}((Ae)^{\oplus n})^{\text{op}}$  is a generating idempotent.

*Proof.* (2) $\Rightarrow$ (1): By Thm. 7.11,  $M$  generates  $G^{\oplus n}$ . So  $M$  is a generator. Since  $G^{\oplus n}p$  is a direct summand of the projective coherent module  $G^{\oplus n}$ ,  $G^{\oplus n}p$  is also projective and coherent.

(1) $\Rightarrow$ (2):  $M$  has an epimorphism from  $G^{\oplus n}$  for some  $n \in \mathbb{Z}_+$ . Since  $M$  is projective, this epimorphism splits. So  $M$  can be viewed as a direct summand of  $G^{\oplus n}$ . Let  $p$  be the projection of  $G^{\oplus n}$  onto  $M$ , which can be viewed as an endomorphism of  $G^{\oplus n}$ . So  $p$  is an idempotent of  $B$ , and  $M = G^{\oplus n}p$ . Since  $M$  is a generator, it generates  $G^{\oplus n}$ . Since  $G^{\oplus n}$  is projective, by Thm. 7.11,  $p$  is generating.  $\square$

## 8 Right pseudotraces

Let  $A$  be an AUF algebra. Let  $B$  be a unital algebra. Let  $M$  be an  $A$ - $B$  bimodule, coherent as a left  $A$ -module.

For each  $y \in B$  and  $\xi \in M$ , we write  $\xi y$  as  $y^{\text{op}}\xi$ . Namely,  $y^{\text{op}}$  is viewed as an element of  $\text{End}_{A,-}(M)$ .

**Definition 8.1.** A **right coordinate system** of  $M$  denotes a collection of morphisms

$$\beta_j \in \text{Hom}_{A,-}(Ae, M) \quad \check{\beta}^j \in \text{Hom}_{A,-}(M, Ae)$$

where  $e \in A$  is an idempotent (called the **domain idempotent**), and  $j$  runs through a *finite* index set  $J$  such that the  $\sum_{j \in J} \beta_j \circ \check{\beta}^j$  equals  $\text{id}_M$ .

**Remark 8.2.**  $M$  has a right coordinate system iff  $M$  is  $A$ -projective.

*Proof.* By Rem. 7.1, each  $N \in \text{Coh}_L(A)$  has an epimorphism from  $(Ae)^{\oplus n}$  where  $e \in A$  is an idempotent and  $n \in \mathbb{Z}_+$ . This epimorphism splits iff  $N$  is projective in  $\text{Coh}_L(A)$ . Therefore, similar to Rem. 4.2, we see that  $M$  has a right coordinate system iff  $M$  is  $A$ -projective.  $\square$

**Remark 8.3.** In Def. 8.1, one can freely enlarge the domain idempotent  $e$ . More precisely, suppose that  $f \in A$  is an idempotent such that  $e \leq f$ . One can define a new right coordinate system

$$\begin{aligned} \gamma_j &\in \text{Hom}_{A,-}(Af, M) & \check{\gamma}^j &\in \text{Hom}_{A,-}(M, Af) \\ \gamma_j(af) &= \beta_j(ae) & \check{\gamma}^j(\xi) &= \check{\beta}^j(\xi) \end{aligned} \tag{8.1}$$

called the **canonical extension** of  $(\beta_j, \check{\beta}^j)_{j \in J}$ .

**Definition 8.4.** Assume that  $M$  has a right coordinate system  $(\beta_j, \check{\beta}^j)_{j \in J}$ . For each  $\psi \in \text{SLF}(A)$ , define the (right) **pseudotrace**  ${}^\psi\text{Tr}$  associated to  $\psi$  to be

$${}^\psi\text{Tr} : B \rightarrow \mathbb{C} \quad {}^\psi\text{Tr}(y) = \sum_{j \in J} \psi((\check{\beta}^j \circ y^{\text{op}} \circ \beta_j)^{\text{op}})$$

noting that  $\check{\beta}^j \circ y^{\text{op}} \circ \beta_j \in \text{End}_{A,-}(Ae) \simeq (eAe)^{\text{op}}$ . In other words,

$$\psi \text{Tr}(y) = \sum_{j \in J} \psi(\check{\beta}^j \circ y^{\text{op}} \circ \beta_j(e)) \quad (8.2)$$

Note that in (8.2) we have  $\beta_j(e) \in M$ , and hence  $\check{\beta}^j \circ y^{\text{op}} \circ \beta_j(e) \in Ae$ . So

$$\check{\beta}^j \circ y^{\text{op}} \circ \beta_j(e) = \check{\beta}^j \circ y^{\text{op}} \circ \beta_j(e^2) = e\check{\beta}^j \circ y^{\text{op}} \circ \beta_j(e) \in eAe$$

**Proposition 8.5.** *Assume that  $M$  is  $A$ -projective. Let  $\psi \in \text{SLF}(A)$ . Then  $\psi \text{Tr} \in \text{SLF}(B)$ . Moreover, the definition of  $\psi \text{Tr}$  is independent of the choice of right coordinate systems.*

*Proof.* From (8.1) and (8.2), it is clear that a canonical extension of the right coordinate system does not affect the value of  $\psi \text{Tr}(y)$ . Also, note that since  $A$  is AUF, for any idempotents  $e_1, e_2 \in A$  there is an idempotent  $e_3$  such that  $e_1, e_2 \leq e_3$ . Therefore, to compare  $\psi \text{Tr}$  defined by two coordinate systems  $(\alpha_\bullet, \check{\alpha}^\bullet)$  and  $(\beta_\bullet, \check{\beta}^\bullet)$ , by performing canonical extensions, it suffices to assume that their domain idempotents are equal. Then one can use the same argument as in Lem. 4.4 to show that  $(\alpha_\bullet, \check{\alpha}^\bullet)$  and  $(\beta_\bullet, \check{\beta}^\bullet)$  define the same  $\psi \text{Tr}$ . Finally, similar to the proof of Prop. 4.5, one shows that  $\psi \text{Tr}$  is symmetric.  $\square$

**Example 8.6.** Let  $M = Ae$  and  $B = eAe$  where  $e \in A$  is an idempotent. Then the identity map on  $Ae$  gives a right coordinate system. From this, one sees that if  $\psi \in \text{SLF}(A)$  then

$$\psi \text{Tr} = \psi|_{eAe}$$

**Example 8.7.** More generally, let  $M = (Ae)^{\oplus n}$  and  $B = \text{End}_{A,-}(M)^{\text{op}}$ . So  $B = eAe \otimes \mathbb{C}^{n \times n}$ . Let

$$\text{tr} : \mathbb{C}^{n \times n} \rightarrow \mathbb{C}$$

be the standard trace on  $\mathbb{C}^{n \times n}$ . A right coordinate system can be chosen to be the  $n$  canonical embeddings  $Ae \rightarrow (Ae)^{\oplus n}$  and the  $n$  canonical projections  $(Ae)^{\oplus n} \rightarrow Ae$ . Then one easily sees that

$$\psi \text{Tr} = \psi|_{eAe} \otimes \text{tr}$$

**Proposition 8.8.** *Assume that  $M$  is  $A$ -projective. Let  $p \in B$  be an idempotent. Let  $\psi \in \text{SLF}(A)$ . Let  $\psi \text{Tr}_M : B \rightarrow \mathbb{C}$  be the pseudotrace associated to  $M$ . Then the pseudotrace  $\psi \text{Tr}_{Mp} : pBp \rightarrow \mathbb{C}$  associated to the  $A$ -( $pBp$ ) bimodule  $Mp$  is equal to  $\psi \text{Tr}_M|_{pBp}$ , i.e.*

$$\psi \text{Tr}_{Mp} = \psi \text{Tr}_M|_{pBp}$$

*Proof.* Let  $(\beta_\bullet, \check{\beta}^\bullet)$  be a right coordinate system (with domain idempotent  $e \in A$ ) as in Def. 8.1. Then one has a right coordinate system

$$\begin{aligned} \gamma_j &\in \text{Hom}_{A,-}(Ae, Mp) & \check{\gamma}^j &\in \text{Hom}_{A,-}(Mp, Ae) \\ \gamma_j(ae) &= \beta_j(ae)p & \check{\gamma}^j(\xi p) &= \check{\beta}^j(\xi p) \end{aligned}$$

noting that  $Mp \leq M$ , and hence  $\check{\gamma}^j$  is simply the restriction of  $\check{\beta}^j$  to  $Mp$ . Using (8.2) one computes that for each  $y \in B$ ,

$$\begin{aligned} \psi \text{Tr}_{Mp}(pyp) &= \sum_j \psi(\check{\gamma}^j \circ (pyp)^{\text{op}} \circ \gamma_j(e)) = \sum_j \psi(\check{\beta}^j \circ (pyp)^{\text{op}} \circ \beta_j(e)p) \\ &= \sum_j \psi(\check{\beta}^j \circ (pyp)^{\text{op}} \circ p^{\text{op}} \circ \beta_j(e)) = \sum_j \psi(\check{\beta}^j \circ (pyp)^{\text{op}} \circ \beta_j(e)) = \psi \text{Tr}_M(pyp) \end{aligned}$$

□

## 9 Equivalence of left and right pseudotraces

Let  $A, B$  be algebras where  $B$  is unital and finite dimensional.

### 9.1 Preliminary discussion

In this subsection, assume that  $A$  is AUF. We shall consider  $M \in \text{Coh}_L(A)$  such that the left and the right pseudotrace constructions are both available to the  $A$ -( $\text{End}_{A,-}(M)^{\text{op}}$ ) bimodule  $M$ . By Rem. 8.2,  $M$  needs to be assumed  $A$ -projective. One also needs  $M$  to be  $\text{End}_{A,-}(M)^{\text{op}}$ -projective. In fact, these two conditions are precisely what ensure that both left and right pseudotraces can be defined.

**Proposition 9.1.** *Let  $M$  be an  $A$ - $B$  bimodule. Assume that  $M$  is  $A$ -coherent. Then the following are equivalent.*

- (1)  $M$  has a left coordinate system.
- (2)  $M$  is  $B$ -projective.

Although this proposition will not be used in the current note, we include it here as it may be of use in the future.

*Proof.* (1) $\Rightarrow$ (2): See Rem. 4.2.

(2) $\Rightarrow$ (1): Let  $(e_i)_{i \in \mathcal{I}}$  be as in Def. 5.1. By Rem. 7.2, each  $e_i M$  is finite-dimensional. Therefore, the right  $B$ -module  $e_i M$  has an epimorphism from  $B^{\oplus n}$  which splits because  $M$  is  $B$ -projective (and hence  $e_i M$  is projective since  $M = \bigoplus_{i \in \mathcal{I}} e_i M$ ). Therefore, for each  $i \in \mathcal{I}$ , there is a finite left coordinate system  $\alpha_{i,\bullet} \in \text{Hom}_{-,B}(B, e_i M)$  and  $\check{\alpha}^{i,\bullet} \in \text{Hom}_{-,B}(e_i M, B)$ . Let

$$\begin{aligned} \gamma_{i,\bullet} &\in \text{Hom}_{-,B}(B, M) & \check{\gamma}^{i,\bullet} &\in \text{Hom}_{-,B}(M, B) \\ \gamma_{i,\bullet}(b) &= \alpha_{i,\bullet}(b) & \check{\gamma}^{i,\bullet}(\xi) &= \check{\alpha}^{i,\bullet}(e_i \xi) \end{aligned}$$

Then one checks easily that  $(\gamma_{i,\bullet}, \check{\gamma}^{i,\bullet})_{i \in \mathcal{I}}$  is a left coordinate system of  $M$ .

□

## 9.2 Calculation of some left pseudotraces

In this subsection,  $A$  is not assumed to be AUF. Let  $M$  be an  $A$ - $B$  bimodule.

The goal of this subsection is to prepare for the proof of the main Thm. 9.4. The following theorem is dual to Prop. 8.8.

**Theorem 9.2.** *Assume that  $M$  has a left coordinate system. Let  $p \in B$  be a generating idempotent. Then the following are true.*

1. *The  $A$ -( $pBp$ ) bimodule  $Mp$  has a left coordinate system.*
2. *Let  $\phi \in \text{SLF}(B)$ . Then on  $A$ , the pseudotrace associated to  $\phi|_{pBp}$  and  $Mp$  is equal to the pseudotrace associated to  $\phi$  and  $M$ . Namely,*

$$\text{Tr}_{Mp}^{\phi|_{pBp}} = \text{Tr}_M^\phi \quad (9.1)$$

In this theorem, we do not require that  $A$  is AUF.

*Proof.* Choose a left coordinate system for  $M$ :

$$\alpha_i \in \text{Hom}_{-,B}(B, M) \quad \check{\alpha}^i \in \text{Hom}_{-,B}(M, B) \quad i \in \mathfrak{I}$$

Since  $p$  is generating, similar to the proof of Thm. 6.5, we can find finitely many elements  $u_k, v_k$  in  $B$  such that

$$\begin{aligned} v_k u_k &= p_k & u_k v_k &= q_k \\ u_k &\in q_k B p_k & v_k &\in p_k B q_k \end{aligned}$$

where  $p_k, q_k \in B$  are idempotents,  $1_B = \sum_k q_k$  is a primitive orthogonal decomposition of  $1_B$ , and  $p_k \leq p$  for each  $k$ .<sup>3</sup> Let

$$\begin{aligned} \theta_{i,k} &\in \text{Hom}_{-,pBp}(pBp, Mp) & \check{\theta}^{i,k} &\in \text{Hom}_{-,pBp}(Mp, pBp) \\ \theta_{i,k}(pyp) &= \alpha_i(u_k \cdot pyp) & \check{\theta}^{i,k}(\xi p) &= v_k \cdot \check{\alpha}^i(\xi p) \end{aligned}$$

noting that  $\alpha_i(u_k \cdot pyp) = \alpha_i(u_k)pyp \in Mp$  and  $v_k \cdot \check{\alpha}^i(\xi p) = v_k \cdot \check{\alpha}^i(\xi)p \in p_k B p \subset pBp$ .

For each  $\xi \in M$ , note that if  $\check{\alpha}^i(\xi) = 0$ , then  $\check{\theta}^{i,k}(\xi p) = v_k \check{\alpha}^i(\xi)p = 0$ . Therefore,  $\check{\theta}^{i,k}(\xi p) = 0$  for all but finitely many  $i$  and  $k$ . Moreover, we compute

$$\begin{aligned} \sum_{i,k} \theta_{i,k} \circ \check{\theta}^{i,k}(\xi p) &= \sum_{i,k} \theta_{i,k}(v_k \check{\alpha}^i(\xi p)) = \sum_{i,k} \alpha_i(u_k v_k \check{\alpha}^i(\xi p)) \\ &= \sum_{i,k} \alpha_i(q_k \check{\alpha}^i(\xi p)) = \sum_i \alpha_i \circ \check{\alpha}^i(\xi p) = \xi p \end{aligned}$$

where all the sums are finite. This proves that  $(\theta, \check{\theta})$  satisfies Def. 4.1-(a). It is easy to check Def. 4.1-(b). So we have proved that  $(\theta, \check{\theta})$  is a left coordinate system of  $Mp$ .

<sup>3</sup>So  $p_k, q_k$  are similar to  $\varepsilon_{k_i}, e_i$  in the proof of Thm. 6.5.

It remains to check (9.1). Choose any  $x \in A$ . By (4.3) and the fact that  $1_{pBp} = p$ ,

$$\begin{aligned} \mathrm{Tr}_{Mp}^{\phi|_{pBp}}(x) &= \sum_{i,k} \phi(\check{\theta}^{i,k} \circ x \circ \theta_{i,k}(p)) = \sum_{i,k} \phi(\check{\theta}^{i,k} \circ x \circ \alpha_i(u_k p)) \\ &= \sum_{i,k} \phi(\check{\theta}^{i,k} \circ x \circ \alpha_i(u_k)) = \sum_{i,k} \phi(v_k \cdot \check{\alpha}^i(x \circ \alpha_i(u_k))) \end{aligned}$$

Since  $\check{\alpha}^i, x, \alpha_i$  commute with the right multiplication by  $v_k$ , and since  $\phi$  is symmetric,

$$\begin{aligned} \mathrm{Tr}_{Mp}^{\phi|_{pBp}}(x) &= \sum_{i,k} \phi(\check{\alpha}^i(x \circ \alpha_i(u_k))v_k) = \sum_{i,k} \phi(\check{\alpha}^i(x \circ \alpha_i(u_k v_k))) \\ &= \sum_i \phi(\check{\alpha}^i(x \circ \alpha_i(1_B))) = \mathrm{Tr}_M^\phi(x) \end{aligned}$$

This finishes the proof of (9.1).  $\square$

**Corollary 9.3.** *Assume that  $M$  has a left coordinate system. Let  $n \in \mathbb{Z}_+$ . Let  $\tilde{B} = B \otimes \mathbb{C}^{n \times n}$ . Then the  $A$ - $\tilde{B}$  bimodule  $M^{\oplus n} \simeq M \otimes \mathbb{C}^{1,n}$  has a left coordinate system. Moreover, for each  $\phi \in \mathrm{SLF}(B)$ , we have*

$$\mathrm{Tr}_{M^{\oplus n}}^{\phi \otimes \mathrm{tr}} = \mathrm{Tr}_M^\phi \quad (9.2)$$

as pseudotraces on  $A$  associated to  $\phi \otimes \mathrm{tr} \in \mathrm{SLF}(\tilde{B})$  and  $\phi \in \mathrm{SLF}(B)$ , respectively.

Recall that  $\mathrm{tr} \in \mathrm{SLF}(\mathbb{C}^{n \times n})$  is the standard trace on the  $n \times n$  matrix algebra.

*Proof.* Choose a left coordinate system

$$\alpha_i \in \mathrm{Hom}_{-,B}(B, M) \quad \check{\alpha}^i \in \mathrm{Hom}_{-,B}(M, B)$$

of  $M$ . Define

$$\gamma_i \in \mathrm{Hom}_{-,\tilde{B}}(\tilde{B}, M^{\oplus n}) \quad \check{\gamma}^i \in \mathrm{Hom}_{-,\tilde{B}}(M^{\oplus n}, \tilde{B})$$

such that

$$\begin{aligned} \gamma_i \begin{bmatrix} y_{1,1} & \cdots & y_{1,n} \\ \vdots & & \\ y_{n,1} & \cdots & y_{n,n} \end{bmatrix} &= [\alpha_i(1_B), 0, \dots, 0] \begin{bmatrix} y_{1,1} & \cdots & y_{1,n} \\ \vdots & & \\ y_{n,1} & \cdots & y_{n,n} \end{bmatrix} = [\alpha_i(y_{1,1}), \dots, \alpha_i(y_{1,n})] \\ \check{\gamma}^i[\xi_1, \dots, \xi_n] &= \begin{bmatrix} \check{\alpha}^i(\xi_1) & \cdots & \check{\alpha}^i(\xi_n) \\ 0 & \cdots & 0 \\ \vdots & \vdots & \vdots \\ 0 & \cdots & 0 \end{bmatrix} \end{aligned}$$

One checks easily that this is a left coordinate system of  $M^{\oplus n}$ . Now (9.2) follows by applying Thm. 9.2 to the  $A$ - $\tilde{B}$  bimodule  $M^{\oplus n}$  and the generating projection  $p \in \tilde{B}$ , where  $p$  is the matrix whose  $(1, 1)$ -entry is 1 and other entries are 0.  $\square$



### 9.3 The main theorem

Assume that  $A$  is strongly AUF (cf. Cor. 6.4) so that  $A$  has a projective generator (cf. Prop. 7.8). The following generalization of Thm. 6.6 is the main theorem of this note.

**Theorem 9.4.** *Assume that  $M \in \text{Coh}_L(A)$  is a projective generator. Assume that  $B = \text{End}_{A,-}(M)^{\text{op}}$  so that  $M$  is an  $A$ - $B$  bimodule. Then  $M$  has left and right coordinate systems. Moreover, we have a linear isomorphism*

$$\text{SLF}(A) \xrightarrow{\sim} \text{SLF}(B) \quad \psi \mapsto \psi \text{Tr} \quad (9.3a)$$

whose inverse map is

$$\text{SLF}(B) \xrightarrow{\sim} \text{SLF}(A) \quad \phi \mapsto \text{Tr}^\phi \quad (9.3b)$$

Of course, both pseudotraces are associated to  $M$ ; we have suppressed the subscript  $M$ .

*Proof.* Note that  $\dim B < +\infty$  by Prop. 7.9. So  $\dim \text{SLF}(B) < +\infty$ . Since  $M \in \text{Coh}_L(A)$  is  $A$ -projective, by Rem. 8.2,  $M$  has a right coordinate system. By Cor. 7.12, we may assume that  $M = G \cdot p$  where

- $G = (Ae)^{\oplus n}$  for some  $n \in \mathbb{Z}_+$  and a generating idempotent  $e \in A$ .
- $M = Gp$  where  $p$  is a generating idempotent of  $\tilde{B} = \text{End}_{A,-}(G)^{\text{op}} = eAe \otimes \mathbb{C}^{n \times n}$ .
- $B = p\tilde{B}p$  (by Prop. 7.10).

By Thm. 6.5 and Cor. 9.3,  $G$  has a left coordinate system. Therefore, by Thm. 9.2,  $M$  has a left coordinate system.

By Thm. 6.6, we have  $\dim \text{SLF}(A) = \dim \text{SLF}(eAe)$ . Clearly we have a linear isomorphism

$$\text{SLF}(eAe) \xrightarrow{\sim} \text{SLF}(eAe \otimes \mathbb{C}^{n \times n}) \quad \omega \mapsto \omega \otimes \text{tr}$$

So  $\dim \text{SLF}(eAe) = \dim \text{SLF}(\tilde{B})$ . By Thm. 6.6, we have  $\dim \text{SLF}(\tilde{B}) = \dim \text{SLF}(B)$ . This proves  $\dim \text{SLF}(A) = \dim \text{SLF}(B) < +\infty$ .

Choose any  $\psi \in \text{SLF}(A)$ . By Exp. 8.7,  ${}^\psi \text{Tr}_G : \tilde{B} \rightarrow \mathbb{C}$  equals  $\psi|_{eAe} \otimes \text{tr}$ . By Prop. 8.8, on  $B = p(eAe \otimes \mathbb{C}^{n \times n})p$  we have

$${}^\psi \text{Tr}_M = (\psi|_{eAe} \otimes \text{tr})|_B =: \phi$$

Now  $\phi \in \text{SLF}(B)$ . By Thm. 9.2 and Cor. 9.3,

$$\text{Tr}_M^\phi = \text{Tr}_{Gp}^{(\psi|_{eAe} \otimes \text{tr})|_B} = \text{Tr}_G^{\psi|_{eAe} \otimes \text{tr}} = \text{Tr}_{Ae}^{\psi|_{eAe}}$$

By Thm. 6.6,  $\text{Tr}_{Ae}^{\psi|_{eAe}} = \psi$ . So  $\text{Tr}_M^\phi = \psi$ . We have thus proved that  $(9.3b) \circ (9.3a)$  is the identity map on  $\text{SLF}(A)$ . This finishes the proof.  $\square$

## 10 Equivalence of non-degeneracy of left and right pseudotraces

**Definition 10.1.** Let  $A$  be an algebra and  $\psi \in \text{SLF}(A)$ . We say that  $\psi$  is **non-degenerate** if

$$\{x \in A : \psi(xA) = 0\} \equiv \{x \in A : \psi(xy) = 0, \forall y \in A\}$$

is zero.

In the following,  $A$  is always assumed to be AUF.

**Lemma 10.2.** *Let  $e \in A$  be an idempotent, and let  $\psi \in \text{SLF}(A)$ . If  $\psi$  is non-degenerate, then the restriction  $\psi|_{eAe}$  is non-degenerate. Conversely, if  $\psi|_{eAe}$  is non-degenerate and  $e$  is generating, then  $\psi$  is non-degenerate.*

*Proof.* Assume that  $\psi$  is non-degenerate. Choose  $x \in eAe$  such that  $\psi(xeAe) = 0$ . Then

$$\psi(xA) = \psi(exeA) = \psi(xeAe) = 0$$

and hence  $x = 0$ . Therefore  $\psi|_{eAe}$  is non-degenerate.

Conversely, assume that  $\psi|_{eAe}$  is non-degenerate and  $e$  is generating. Choose  $x \in A$  such that  $\psi(xA) = 0$ . Then for each  $a, b \in A$ ,

$$\psi(eaxbe \cdot eAe) = \psi(eaxbeAe) = \psi(xbeAea) = 0$$

Therefore  $eaxbe = 0$ . Since  $b$  is arbitrary, we have  $eaxAe = 0$ . Since  $e$  is generating, it is not hard to show that the left  $A$ -module  $Ae$  is faithful. (See for example Lem. 11.6.) It follows that  $eax = 0$ . Therefore  $eAx = 0$ . Similarly,  $eA$  is a faithful right  $A$ -module. Hence  $x = 0$ . This proves the non-degeneracy of  $\psi$ .  $\square$

**Proposition 10.3.** *Assume that  $\psi \in \text{SLF}(A)$  is non-degenerate. Let  $M \in \text{Coh}_L(A)$  be projective, and let  $B = \text{End}_{A,-}(M)^{\text{op}}$ . Then the right pseudotrace  ${}^\psi\text{Tr} \in \text{SLF}(B)$  is non-degenerate.*

*Proof.* By Prop. 2.6,  $M$  can be viewed as a direct summand of  $\bigoplus_{i=1}^n Ae_i$  where each  $e_i \in A$  is an idempotent. Let  $e \in A$  be an idempotent such that  $e \geq e_i$  for all  $i$ . Then  $M$  is a direct summand of  $(Ae)^{\oplus n}$ . By Prop. 1.2, we have  $\text{End}_{A,-}(Ae)^{\text{op}} = eAe$ , and hence  $\text{End}_{A,-}((Ae)^{\oplus n})^{\text{op}} = eAe \otimes \mathbb{C}^{n \times n}$ . By Prop. 7.10, there is an idempotent  $p \in eAe \otimes \mathbb{C}^{n \times n}$  such that  $M = (Ae)^{\oplus n}p$ . By Lem. 10.2,  $\psi|_{eAe}$  is non-degenerate, and hence  $\psi|_{eAe} \otimes \text{tr} : eAe \otimes \mathbb{C}^{n \times n} \rightarrow \mathbb{C}$  is non-degenerate. By Lem. 10.2 again, the restriction of  $\psi|_{eAe} \otimes \text{tr}$  to  $p(eAe \otimes \mathbb{C}^{n \times n})p$  (which is  $B$  due to Prop. 7.10) is non-degenerate. But this restriction is exactly  ${}^\psi\text{Tr}$  due to Exp. 8.7 and Prop. 8.8.  $\square$

**Theorem 10.4.** *Assume that  $A$  is strongly AUF. Then in Thm. 9.4, for any  $\psi \in \text{SLF}(A)$ , the non-degeneracy of  $\psi$  and of  ${}^\psi\text{Tr}$  are equivalent.*

*Proof.* We use the notation in the proof of Thm. 9.4. From that proof, we know  ${}^\psi\text{Tr} = (\psi|_{eAe} \otimes \text{tr})|_B$ . By Lem. 10.2,  $\psi$  is non-degenerate iff  $\psi|_{eAe}$  is so, and  $\psi|_{eAe} \otimes \text{tr}$  is non-degenerate iff  $(\psi|_{eAe} \otimes \text{tr})|_B$  is so. The equivalence of the non-degeneracy of  $\psi|_{eAe}$  and of  $\psi|_{eAe} \otimes \text{tr}$  is obvious. The proof is finished.  $\square$

## 11 Classification of strongly AUF algebras

In this section, we fix an AUF algebra  $A$ .

**Definition 11.1.** For each left  $A$ -module  $M$ , let  $M^*$  be the space of linear functionals, which has a right  $A$ -module structure defined by

$$(\phi a)(m) = \phi(am) \quad \text{for all } a \in A, m \in M$$

We define the **quasicoherent dual**

$$\begin{aligned} M^\vee &= \{\phi \in M^* : \phi \in \phi \cdot A\} \\ &= \{\phi \in M^* : \text{there exists an idempotent } e \in A \text{ such that } \phi = \phi e\} \end{aligned}$$

By Def. 2.2,  $M^\vee$  is the largest right  $A$ -submodule of  $M^*$  that is quasicoherent.

**Remark 11.2.** Let  $M \in \text{QCoh}_L(A)$ . Let  $(e_i)_{i \in \mathfrak{I}}$  be as in Def. 5.1. Then, as vector spaces, we clearly have

$$M = \bigoplus_{i \in \mathfrak{I}} e_i M \quad M^* = \prod_{i \in \mathfrak{I}} (e_i M)^*$$

It follows easily that

$$M^\vee = \bigoplus_{i \in \mathfrak{I}} (e_i M)^*$$

**Definition 11.3.** For each  $M \in \text{QCoh}_L(A)$ , we let

$$\text{End}^0(M) = M \otimes_{\mathbb{C}} M^\vee$$

viewed as a subalgebra of  $\text{End}(M)$ .<sup>4</sup> If  $B$  is an algebra, and  $M$  has a right  $B$ -module structure commuting with the left action of  $A$ , we let

$$\text{End}_{-,B}^0(M) = \{T \in \text{End}^0(M) : (T\xi)b = T(\xi b) \text{ for all } \xi \in M, b \in B\} \quad (11.1)$$

**Remark 11.4.** Let  $M \in \text{Coh}_L(A)$ . By Rem. 7.2 we have  $\dim e_i M < +\infty$ . It follows from Rem. 11.2 that

$$\text{End}^0(M) = \{T \in \text{End}(M) : Te_i = 0 \text{ for all but finitely many } i \in \mathfrak{I}\}$$

**Proposition 11.5.** Choose  $M \in \text{Coh}_L(A)$ , and let  $B = \text{End}_{A,-}(M)^{\text{op}}$ . Then for each generating idempotent  $p \in B$ , we have a linear isomorphism

$$\text{End}_{-,B}^0(M) \xrightarrow{\cong} \text{End}_{-,pBp}^0(Mp) \quad S \mapsto S|_{Mp} \quad (11.2)$$

---

<sup>4</sup>That is, for each  $\xi \in M$ ,  $\phi \in M^\vee$ , the operator  $\xi \otimes \phi$  sends each  $\eta \in M$  to  $\phi(\eta) \cdot \xi$ .

*Proof.* Step 1. Let  $\hat{B} = B^{\text{op}} = \text{End}_{A,-}(M)$ , and let  $\hat{p} \in \hat{B}$  be the opposite element of  $p$ . Then  $M$  has a left  $\hat{B}$ -module structure commuting with the left action of  $A$ , and  $R_p$  is the left multiplication by  $\hat{p}$ .

For each  $S \in \text{End}_{-,B}^0(M)$ , note that  $S|_{Mp} = S|_{\hat{p}M}$  maps  $\hat{p}M$  into  $\hat{p}M$ , because  $S\hat{p}\xi = \hat{p}S\xi \in \hat{p}M$  for each  $\xi \in M$ . It is clear that  $S|_{Mp}$  commutes with the action of  $\hat{p}\hat{B}\hat{p}$ . It can be checked from Rem. 11.4 that  $S|_{Mp}$  belongs to  $\text{End}^0(M)$ . This proves that  $S|_{Mp}$  belongs to  $\text{End}_{-,pBp}^0(Mp)$ . We have thus proved that the linear map (11.2) is well-defined.

Step 2. Let us prove the surjectivity of (11.2). By Prop. 7.9,  $B$  is finite-dimensional. Therefore, we have an orthogonal primitive decomposition  $1_{\hat{B}} - \hat{p} = f_1 + \cdots + f_n$  in  $\hat{B}$ . In this case, we have

$$M = \hat{p}M \oplus f_1M \oplus \cdots \oplus f_nM$$

By Prop. 6.2, for each  $1 \leq i \leq n$ ,  $f_i$  is equivalent to a sub-idempotent  $q_i$  of  $\hat{p}$ , i.e., there exist  $u_i \in f_i\hat{B}q_i$  and  $v_i \in q_i\hat{B}f_i$  such that  $u_iv_i = f_i$  and  $v_iu_i = q_i \leq \hat{p}$  (where  $q_i \in \hat{B}$  is an idempotent).

Now, we choose  $T \in \text{End}_{-,pBp}^0(Mp) = \text{End}_{-,pBp}^0(\hat{p}M)$ . Define a linear map

$$S : M \rightarrow M \quad \xi \mapsto T(\hat{p}\xi) + \sum_{i=1}^n u_i T(v_i \xi) \quad (11.3)$$

By Rem. 11.4, we have  $S \in \text{End}^0(M)$ . We claim that  $S$  commutes with the action of  $\hat{B}$  (and hence  $S \in \text{End}_{-,B}^0(M)$ ). If this is proved, then since  $T$  clearly equals  $S|_{Mp} = S|_{\hat{p}M}$  (because  $v_i\hat{p} = 0$ , see below), the proof of the surjectivity of (11.2) is complete.

Note that since  $\hat{p}, f_1, \dots, f_n$  are mutually orthogonal, we have

$$\begin{aligned} u_i u_j &= 0 & v_i v_j &= 0 & \text{for all } i, j \\ v_j u_i &= 0 & & & \text{for all } i \neq j \\ v_i \hat{p} &= 0 & \hat{p} u_i &= 0 & \text{for all } i \end{aligned}$$

Using this observation and the fact that  $T : \hat{p}M \rightarrow \hat{p}M$  commutes with the left action of  $\hat{p}\hat{B}\hat{p}$ , we compute that for each  $j$  and  $\xi \in M$ ,

$$\begin{aligned} S(v_j \xi) &= T(\hat{p}v_j \xi) + 0 = T(v_j \xi) \\ v_j S(\xi) &= v_j T(\hat{p}\xi) + v_j u_j T(v_j \xi) \stackrel{v_j u_j = q_j \in \hat{p}\hat{B}\hat{p}}{=} 0 + T(q_j v_j \xi) = T(v_j \xi) \end{aligned}$$

and hence  $S(v_j \xi) = v_j S(\xi)$ ; similarly,

$$\begin{aligned} S(u_j \xi) &= T(\hat{p}u_j \xi) + u_j T(v_j u_j \xi) \stackrel{v_j u_j = q_j \in \hat{p}\hat{B}\hat{p}}{=} 0 + u_j q_j T(\hat{p}\xi) = u_j T(\hat{p}\xi) \\ u_j S(\xi) &= u_j T(\hat{p}\xi) + 0 = u_j T(\hat{p}\xi) \end{aligned}$$

and hence  $S(u_j \xi) = u_j S(\xi)$ . Moreover, for each  $b \in \hat{B}$  we have

$$S(\hat{p}b\hat{p}\xi) = T(\hat{p}b\hat{p}\xi) + 0 = \hat{p}b\hat{p}T(\hat{p}\xi)$$

$$\hat{p}b\hat{p}S(\xi) = \hat{p}b\hat{p}T(\hat{p}\xi) + 0 = \hat{p}b\hat{p}T(\hat{p}\xi)$$

and hence  $S(\hat{p}b\hat{p}\xi) = \hat{p}b\hat{p}S(\xi)$ . This proves that  $S$  commutes with the left action of  $\hat{B}$ , since  $\hat{B}$  is generated by  $\{u_i, v_i : 1 \leq i \leq n\}$  and  $\hat{p}\hat{B}\hat{p}$ —to see this, note that for each  $b \in \hat{B}$ , by setting  $f_0 = u_0 = v_0 = \hat{p}$ , we have

$$b = \sum_{i,j=0}^n f_i b f_j = \sum_{i,j=0}^n u_i b_{i,j} v_j$$

where each  $b_{i,j} := v_i b u_j$  commutes with the left actions of  $A$  and satisfies  $b_{i,j} = \hat{p}b_{i,j}\hat{p}$ , and hence belongs to  $\hat{p}\hat{B}\hat{p}$ .

Step 3. If  $S \in \text{End}_{-,B}^0(M)$  and  $S|_{\hat{p}M} = 0$ , then for each  $\xi \in M$ , we have

$$S(\xi) = S(\hat{p}\xi) + \sum_{i=1}^n S(f_i\xi) = S(\hat{p}\xi) + \sum_{i=1}^n u_i S(v_i\xi)$$

where  $\hat{p}\xi, v_j\xi \in \hat{p}M$ . Therefore  $S = 0$ . This proves that (11.2) is injective.  $\square$

**Lemma 11.6.** *Suppose that  $e \in A$  is a generating idempotent. Then we have a linear isomorphism*

$$A \xrightarrow{\cong} \text{End}_{-,eAe}^0(Ae) \quad (11.4)$$

*sending each  $a \in A$  to the left multiplication by  $a$ .*

*Proof.* It is obvious that the left action on  $Ae$  by  $a \in A$  belongs to  $\text{End}_{-,eAe}^0(Ae)$ . Therefore, the map (11.4) is well-defined.

Suppose that the left multiplication of  $a \in A$  on  $Ae$  is zero. Then  $aAe = 0$ . Since  $A$  is AUF and hence almost unital, there is an idempotent  $p \in A$  such that  $a = ap$ . Since  $e$  is generating, by Cor. 7.7,  $Ae$  is a generator of  $\text{Coh}_L(A)$ . Therefore,  $Ap$  is a quotient module of  $(Ae)^{\oplus n}$  for some  $n \in \mathbb{Z}_+$ . Thus  $aAp$  is a quotient space of  $(aAe)^{\oplus n}$ , and hence  $aAp = 0$ . This proves  $ap = 0$ , and hence  $a = 0$ . We have thus proved that (11.4) is injective.

Choose  $T \in \text{End}_{-,eAe}^0(Ae)$ . Since  $T \in \text{End}^0(Ae)$ , by Rem. 11.2, there is an idempotent  $f \in A$  such that  $T = fTf$ . It follows that  $fTf|_{fAe}$  belongs to  $\text{End}_{-,eAe}(fAe)$ . Since  $A$  is AUF, we may enlarge  $f$  so that  $e \leq f$  also holds. We claim that  $\text{End}_{-,eAe}(fAe)$  consists of the left multiplications by elements of  $fAf$ . If this is true, then  $T|_{fAe} = fTf|_{fAe}$  is the left multiplication by  $faf$  for some  $a \in A$ . It follows that for any  $b \in A$ , we have  $Tbe = Tfbe = fafbe$ , and hence  $T$  is the left multiplication by  $faf$  on  $Ae$ , finishing the proof that (11.4) is surjective.

By Cor. 6.3, the idempotent  $e \in fAf$  is generating in  $fAf$ . Applying Prop. 11.5 to the finite-dimensional unital algebra  $fAf$  and its (finite-dimensional) coherent left module  $fAf$ , we see that  $\text{End}_{-,eAe}(fAe) = fAf|_{fAe}$ . This proves the claim.  $\square$

**Theorem 11.7.** *Suppose that  $A$  is strongly AUF, and let  $G$  be a projective generator of  $\text{Coh}_L(A)$  (which exists due to Prop. 7.8). Set  $B = \text{End}_{A,-}(G)^{\text{op}}$ . Regard  $G$  as an  $A$ - $B$  bimodule. Then we have a linear isomorphism*

$$A \xrightarrow{\cong} \text{End}_{-,B}^0(G) \quad (11.5)$$

*sending each  $a \in A$  to the left multiplication of  $a$  on  $G$ .*

*Proof.* By Cor. 6.4 and Prop. 7.8,  $A$  has a generating idempotent  $e$ . If  $G = Ae$ , then  $\text{End}_{A,-}(G)^{\text{op}} = eAe$  due to Prop. 1.2. Therefore, by Lem. 11.6, the map (11.5) is bijective.

If  $G = (Ae)^{\oplus n}$  where  $n \in \mathbb{Z}_+$ , one easily checks that  $B = eAe \otimes \mathbb{C}^{n \times n}$  where  $\mathbb{C}^{n \times n}$  is the matrix algebra of order  $n$ . The bijectivity of (11.5) then follows easily.

Finally, let  $G$  be any general projective generator. By Cor. 7.12, we may assume that  $G = (Ae)^{\oplus n}p$  where  $n \in \mathbb{Z}_+$ , and  $p$  is a generating idempotent of  $\tilde{B} = \text{End}_{A,-}((Ae)^{\oplus n})^{\text{op}} \simeq eAe \otimes \mathbb{C}^{n \times n}$ . By Prop. 7.10, we have  $B = p\tilde{B}p$ . Therefore, by Prop. 11.5, the map

$$\text{End}_{-, \tilde{B}}^0((Ae)^{\oplus n}) \rightarrow \text{End}_{-, B}^0(G)$$

sending each  $S$  to  $S|_G$  is bijective. By the previous paragraph, the map

$$A \rightarrow \text{End}_{-, \tilde{B}}^0((Ae)^{\oplus n})$$

sending each  $a$  to the left multiplication by  $a$  is bijective. Therefore, their composition, namely (11.5), is bijective.  $\square$

**Remark 11.8.** In Thm. 11.7, the right  $B$ -module  $G$  is a **projective generator** in the category  $\text{Mod}^R(B)$  of right  $B$ -modules—that is,  $G$  is projective in  $\text{Mod}^R(B)$ , and any object in  $\text{Mod}^R(B)$  has an epimorphism from a (possibly infinite) direct sum of  $G$ .

*Proof.* The projectivity of  $G$  in  $\text{Mod}^R(B)$  is due to Thm. 9.4 and Rem. 4.2. Using the notation in the proof of Thm. 11.7, we may assume  $G = (Ae)^{\oplus n}p$  and  $B = p(eAe \otimes \mathbb{C}^{n \times n})p$  where  $e \in A$  and  $p \in eAe \otimes \mathbb{C}^{n \times n}$  are generating idempotents. Since  $B$  is unital,  $B$  is generating in  $\text{Mod}^R(B)$ . Therefore  $(eAe \otimes \mathbb{C}^{n \times n})p$  is generating in  $\text{Mod}^R(B)$ . Since  $(eAe \otimes \mathbb{C}^{n \times n})p$  is a direct sum of  $(eAe \otimes \mathbb{C}^{1 \times n}) = (eAe)^{\oplus n}p = eG$ , we conclude that  $eG$  is generating in  $\text{Mod}^R(B)$ . Therefore  $G$  is generating in  $\text{Mod}^R(B)$ .  $\square$

**Theorem 11.9.** *Let  $A$  be an algebra. The following are equivalent.*

- (1)  $A$  is strongly AUF.
- (2)  $\mathcal{A}$  is isomorphic to  $\text{End}_{-, B}^0(M)$  where  $B$  is a unital finite-dimensional algebra,  $M$  is a projective generator in  $\text{Mod}^R(B)$ , the vector space  $M$  has a grading

$$M = \bigoplus_{i \in \mathcal{I}} M(i)$$

where each  $M(i)$  is finite-dimensional and is preserved by the right action of  $B$ , and  $\text{End}_{-, B}^0(M)$  is defined by

$$\begin{aligned} \text{End}_{-, B}^0(M) &:= \{T \in \text{End}(M) : (Tm)b = T(mb) \text{ for all } m \in M, b \in B, \\ &\quad T|_{M(i)} = 0 \text{ for all but finitely many } i \in \mathcal{I}\} \end{aligned}$$

*Proof.* The direction (1) $\Rightarrow$ (2) follows from Thm. 11.7 and Rem. 11.8. Let us prove the other direction.

Assume that  $\mathcal{A} = \text{End}_{-, B}^0(M)$  where  $\text{End}_{-, B}^0(M)$  is described as in (2). Let  $e_i$  be the projection of  $M$  onto  $M(i)$ . Then  $e_i$  clearly belongs to  $\mathcal{A}$ , and each  $T \in \mathcal{A}$  can be written as  $T = \sum_{i, j \in \mathcal{I}} e_i T e_j$  where  $e_i T e_j = 0$  for all but finitely many  $i, j$ . This proves that  $\mathcal{A}$  is AUF.

Since  $M$  is a projective generator in  $\text{Mod}^R(B)$ , for each finite subset  $I \subset \mathfrak{I}$ ,  $M_I := \bigoplus_{i \in I} M(i)$  is projective in  $\text{Mod}^R(B)$  (since it is a direct summand of  $M$ ). Let  $1_B = p_1 + \cdots + p_n$  be an orthogonal primitive decomposition of  $1_B$  in  $B$ . By Thm. 5.5, irreducible finite-dimensional right  $B$ -modules are precisely those that are isomorphic to  $p_k B / \text{rad}(p_k B)$  for some  $k$ . Since  $M$  is generating in  $\text{Mod}^R(B)$ , it has an epimorphism to  $p_k B / \text{rad}(p_k B)$  for each  $k$ . This epimorphism must restrict to a nonzero morphism (and hence an epimorphism)  $M(i_k) \rightarrow p_k B / \text{rad}(p_k B)$ . Let  $I = \{i_1, \dots, i_n\}$ . Then  $M_I$  has an epimorphism to each irreducible right  $B$ -module. It follows from Prop. 7.6 that  $M_I$  is a projective generator in the category of finite-dimensional right  $B$ -modules.

Let  $e_I = \sum_{i \in I} e_i$ , which is an idempotent in  $\mathcal{A}$ . We claim that  $e_I$  is a generating idempotent in  $\mathcal{A}$ , which will complete the proof that  $\mathcal{A}$  is strongly AUF.

Let  $\varepsilon$  be any primitive idempotent of  $\mathcal{A}$ . Then  $\varepsilon M$  is a finite-dimensional right  $B$ -module, since any element of  $\mathcal{A}$  has finite range when acting on  $M$ . Moreover, since  $\varepsilon$  is primitive in  $\mathcal{A}$ , the right  $B$ -module  $\varepsilon M$  is indecomposable. Since  $\varepsilon M$  is a direct summand of the projective right  $B$ -module  $M$ , it follows that  $\varepsilon M$  is a finite-dimensional indecomposable projective right  $B$ -module. Therefore, since  $M_I = e_I M$  is a projective generator, similar to the end of the proof of Thm. 7.11, we conclude that the right  $B$ -module  $\varepsilon M$  is isomorphic to a direct summand of  $e_I M$ . Thus, by Prop. 7.10,  $\varepsilon$  is equivalent to a sub-idempotent of  $e_I$  in  $\mathcal{A}$ . This proves the claim that  $e_I$  is generating.  $\square$

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